COURSE SLIDES · WSQ

Application of BIM using Revit



WSQ Course Code: TGS-2022015370

Skills Framework TSC: Building Information Modelling Application (BEV-TEM-3027-1.1-1)

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v10 · 18 August 2026

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

COURSE ADMINISTRATION

Welcome & Housekeeping

© 2026 Tertiary Infotech Academy Pte Ltd

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

© 2026 Tertiary Infotech Academy Pte Ltd

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

© 2026 Tertiary Infotech Academy Pte Ltd

YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE
BIM and parametric design — Autodesk Revit, AutoCAD, Fusion 360 and engineering CAD/CAE toolchains.

DELIVERS
WSQ courses on BIM with Revit, architecture drawing, technical drawing and product design.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

© 2026 Tertiary Infotech Academy Pte Ltd

ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with CAD, Revit or the built-environment industry (if any).
- What you want to be able to model, manage or submit with BIM after this course.

© 2026 Tertiary Infotech Academy Pte Ltd

HOUSEKEEPING

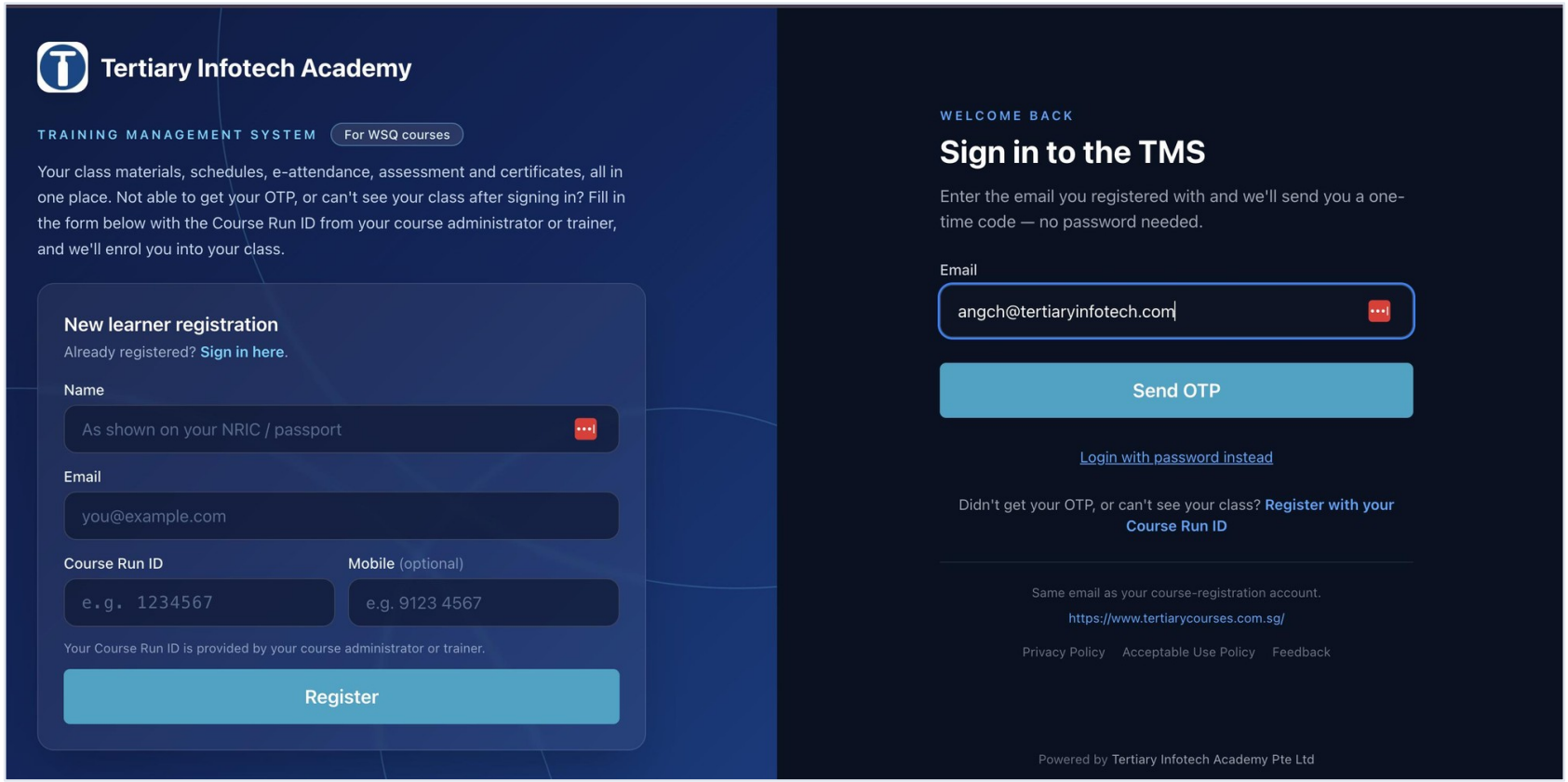
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

© 2026 Tertiary Infotech Academy Pte Ltd

SCHEDULE

Lesson Plan — 2 Days, 8 Hours/Day

Day 1

- Day 1 — BIM fundamentals, Revit and BIM modeling
  - Digital attendance (AM) · welcome & introductions
  - Topic 1: Introduction to BIM and Revit (Labs 1–2)
  - Lunch · digital attendance (PM)
  - Topic 2: BIM Modeling (Labs 3–6)
  - Day 1 recap and Q&A

Day 2

- Day 2 — BIM application, documentation and assessment
  - Digital attendance (AM) · Day 1 recap
  - Topic 3: BIM Application (Labs 7–8)
  - Lunch · digital attendance (PM)
  - Topic 4: BIM Documentation (Labs 9–10)
  - Final Assessment: WA (SAQ) + Practical Performance
  - Daily timing: 9:30am–6:30pm · 1-hour lunch · tea breaks within

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1

Apply Revit for Building Information Modelling (BIM).

2

LO2

Develop models and integrate design for BIM.

3

LO3

Apply BIM and check the output.

4

LO4

Maintain BIM database and develop BIM documentation.

© 2026 Tertiary Infotech Academy Pte Ltd

SKILLS FRAMEWORK ALIGNMENT

Skills Framework for the Built Environment

TSC: Building Information Modelling Application · TSC Code: BEV-TEM-3027-1.1-1

© 2026 Tertiary Infotech Academy Pte Ltd

BEV-TEM-3027-1.1-1 · KNOWLEDGE K1–K10

TSC Knowledge — What the WA (SAQ) Assesses

1. **K1** Principles of BIM
2. **K2** Value proposition of BIM
3. **K3** Requirements of BIM
4. **K4** Definition of BIM
5. **K5** Application of BIM
6. **K6** Technology used in BIM
7. **K7** BIM design processes
8. **K8** Documentation required for BIM
9. **K9** Databases and information systems required for BIM
10. **K10** BIM e-submission documentation requirements and standards

© 2026 Tertiary Infotech Academy Pte Ltd

TSC Abilities — What the Practical Performance Assesses

1. **A1**
   Identify BIM application and development in industry
2. **A2**
   Identify BIM's interoperability with other analysis tools
3. **A3**
   Maintain databases and information systems for BIM
4. **A4**
   Develop models containing building elements and information
5. **A5**
   Integrate the design of active systems in reference models
6. **A6**
   Operate BIM applications and software
7. **A7**
   Interpret data within BIM outputs
8. **A8**
   Analyse the design performance and compliance of the relevant systems
9. **A9**
   Develop BIM documentation

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE MAP

Course Outline — Four Topics, Ten Hands-On Labs

1

Topic 1 · Introduction to BIM and Revit

K1–K4, A1–A2 — BIM principles, dimensions, value, requirements; the Revit environment. Labs 1–2.

2

Topic 2 · BIM Modeling

K6–K7, A4–A5 — walls, floors, roofs, stairs, structure, site, massing, families. Labs 3–6.

3

Topic 3 · BIM Application

K5, A6–A8 — operate BIM, schedules and takeoffs, energy analysis, compliance checks. Labs 7–8.

4

Topic 4 · BIM Documentation

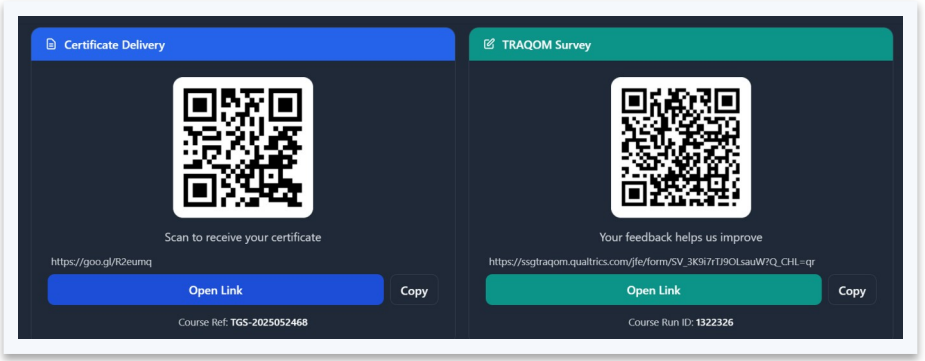
K8–K10, A3, A9 — documentation, databases, worksharing, BCA e-submission standards. Labs 9–10.

© 2026 Tertiary Infotech Academy Pte Ltd

TEST YOURSELF · exams.tertiaryinfotech.com

Practice Exam

- Sharpen your readiness with the Tertiary Infotech practice exams portal.
- Portal: https://exams.tertiaryinfotech.com — search for the BIM / Revit practice exam.
- Attempt it under timed conditions and review every explanation.
- Revisit any topic you miss, then re-take the practice exam before the final assessment.



© 2026 Tertiary Infotech Academy Pte Ltd

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

© 2026 Tertiary Infotech Academy Pte Ltd

FINAL ASSESSMENT · DAY 2

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on Revit modeling tasks, 1.5 hours, open book.
- Format: Open Book — slides, Learner Guide and approved materials only.
- The PP is completed in Revit using the provided PP dataset (.rvt).
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

© 2026 Tertiary Infotech Academy Pte Ltd

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ, 1 hr) then PP (1.5 hrs) — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

© 2026 Tertiary Infotech Academy Pte Ltd

01

TOPIC 01

Introduction to BIM and Revit

What is BIM · principles of BIM · value proposition · requirements and applications · introduction of Revit for BIM

© 2026 Tertiary Infotech Academy Pte Ltd

ALIGNMENT: K1–K4 · A1, A2 · LO1

Key Concepts — Introduction to BIM and Revit

1

BIM is a process, not a drawing

An intelligent 3D model-based process giving AEC professionals the insight to plan, design, construct and manage buildings and infrastructure.

2

One model, whole lifecycle

The BIM process — Plan, Design, Build, Operate — creates intelligent data that is used across the entire life of a built asset.

3

Dimensions beyond 3D

3D geometry, 4D time, 5D cost, 6D sustainability, 7D facility management and 8D safety add information layers to the model.

4

The value proposition

Greater efficiency, lower cost and wastage, better estimates, deeper insight, closer collaboration and better end results.

5

Requirements frame the work

OIR, AIR, EIR, the BIM Execution Plan (BEP), PIM and AIM define what information is delivered, by whom and when.

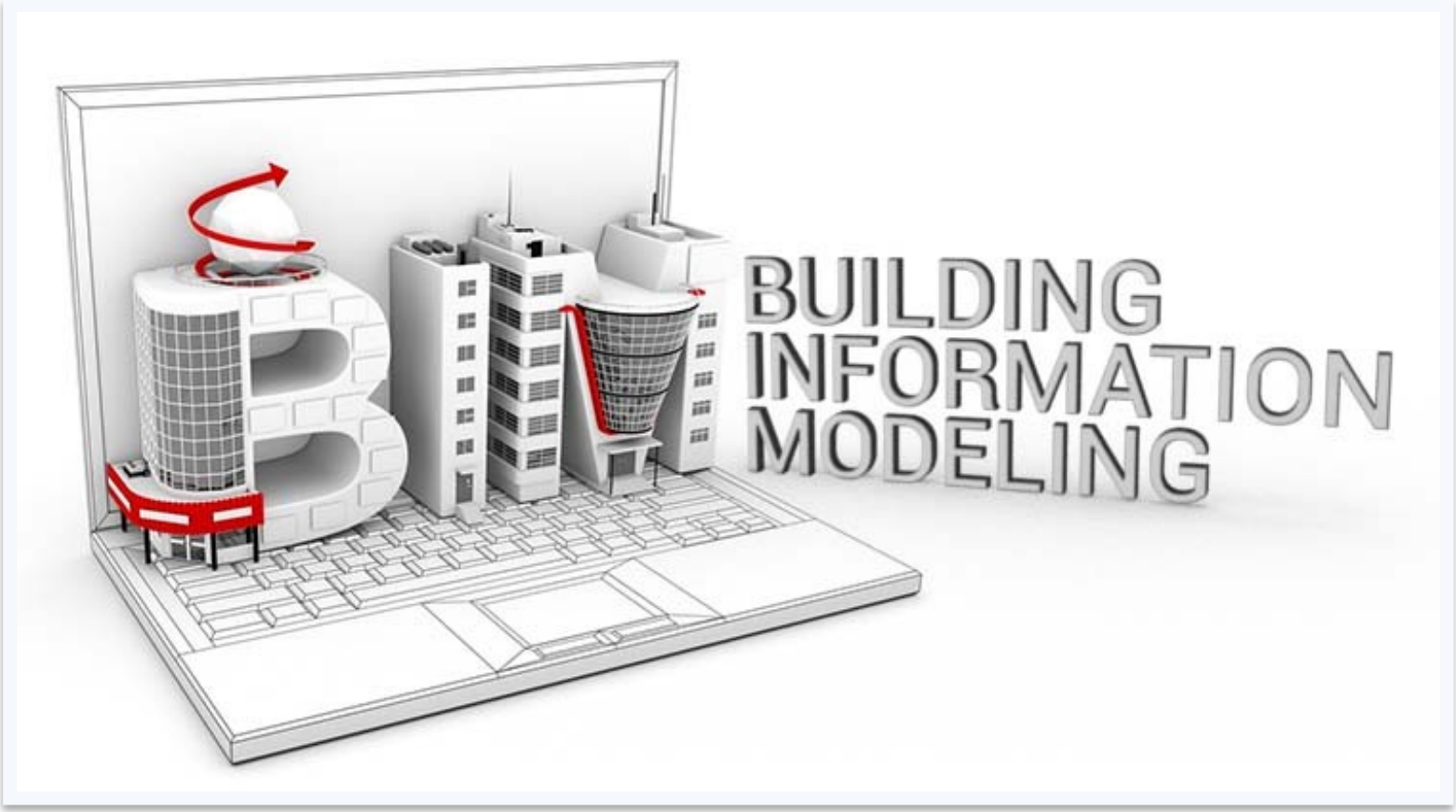
6

Revit is a BIM authoring tool

A single, integrated parametric model to conceptualise, design and document a project — used by architects, engineers and contractors.

© 2026 Tertiary Infotech Academy Pte Ltd

BIM in the Building Lifecycle



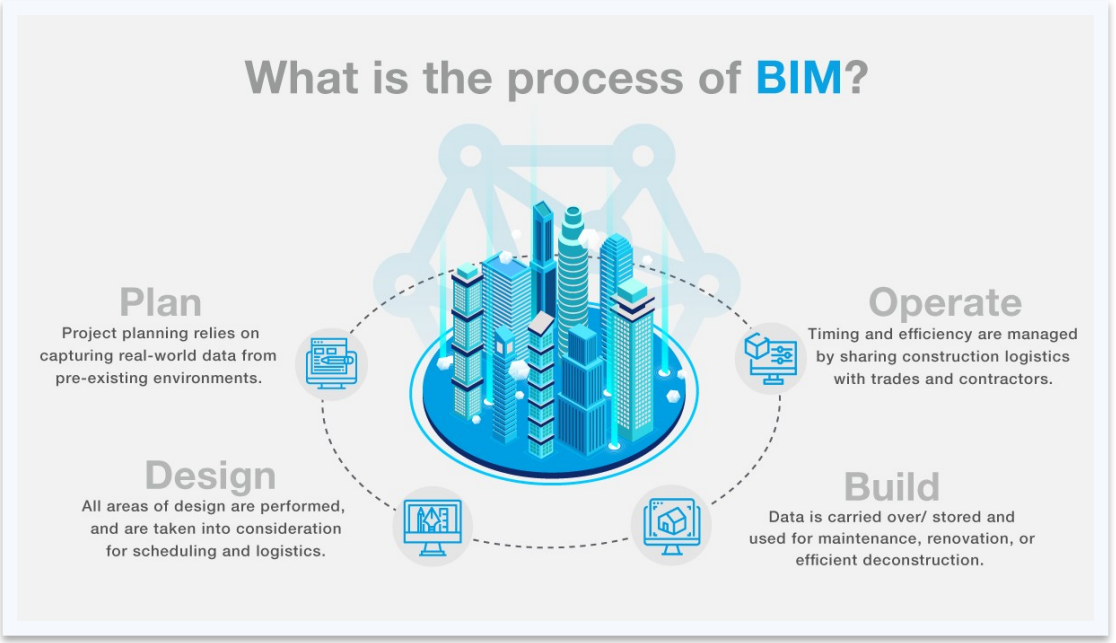
Building Information Modelling spans design, construction and operation.

© 2026 Tertiary Infotech Academy Pte Ltd

DEFINITION · K4

What is BIM?

- An intelligent 3D model-based process that gives AEC professionals every detail needed to plan, design, construct and manage buildings and infrastructure.
- Design and construction teams work more efficiently while capturing the data they create — benefiting operations, maintenance and future planning.
- Used across architecture, civil engineering, construction, plant, MEP and structural engineering to make better design decisions and improve building performance.



© 2026 Tertiary Infotech Academy Pte Ltd

BIM PROCESS · K1

The Process of BIM — Plan · Design · Build · Operate

1

Plan

Combine reality capture and real-world data to generate context models of the existing built and natural environment.

2

Design

Conceptual design, analysis, detailing and documentation — BIM data informs scheduling and logistics before construction begins.

3

Build

Fabrication starts from BIM specifications; construction logistics are shared with trades and contractors for optimum timing.

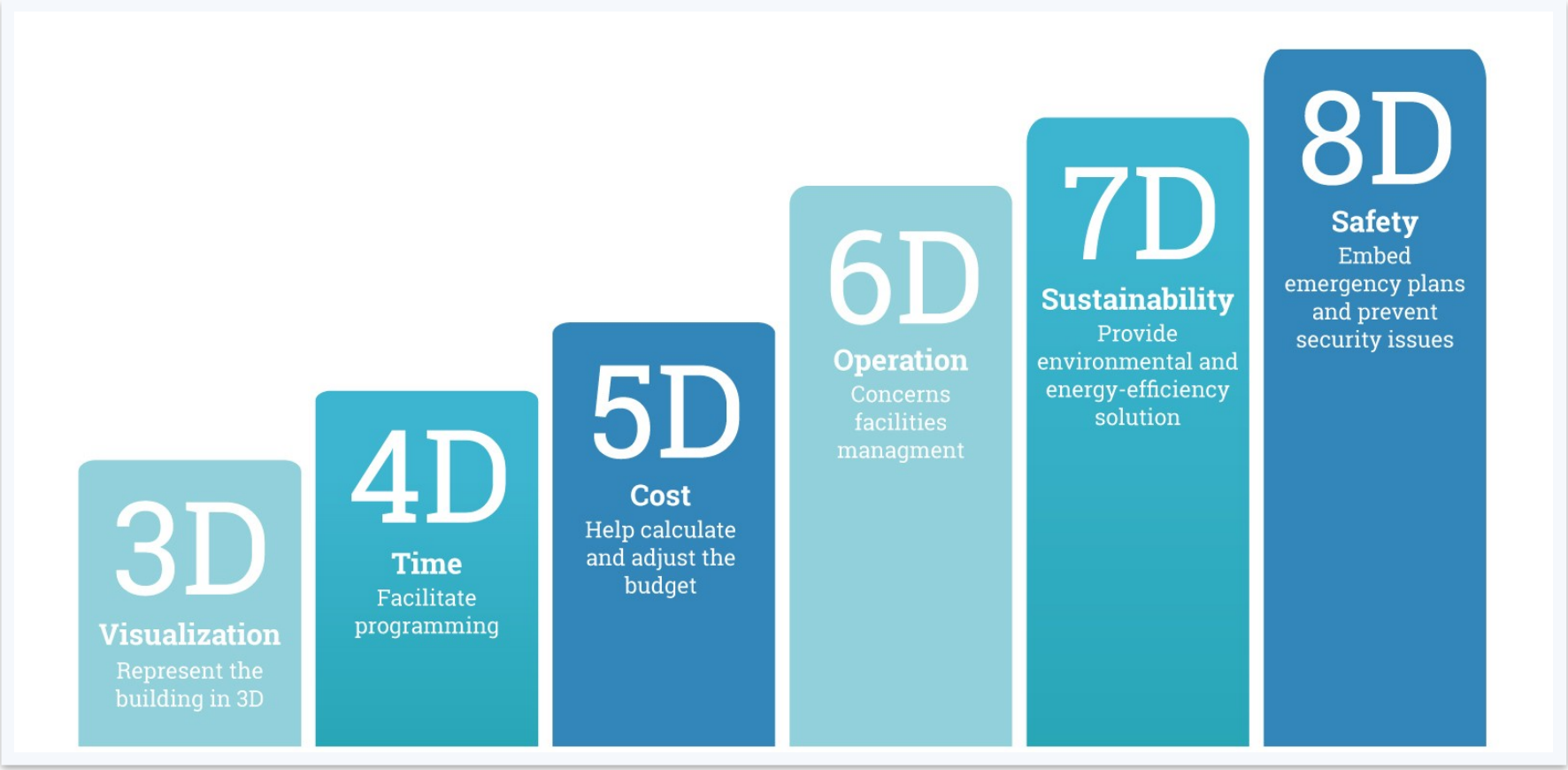
4

Operate

BIM data carries over to operations and maintenance — and later to cost-effective renovation or efficient deconstruction.

© 2026 Tertiary Infotech Academy Pte Ltd

Principles of BIM

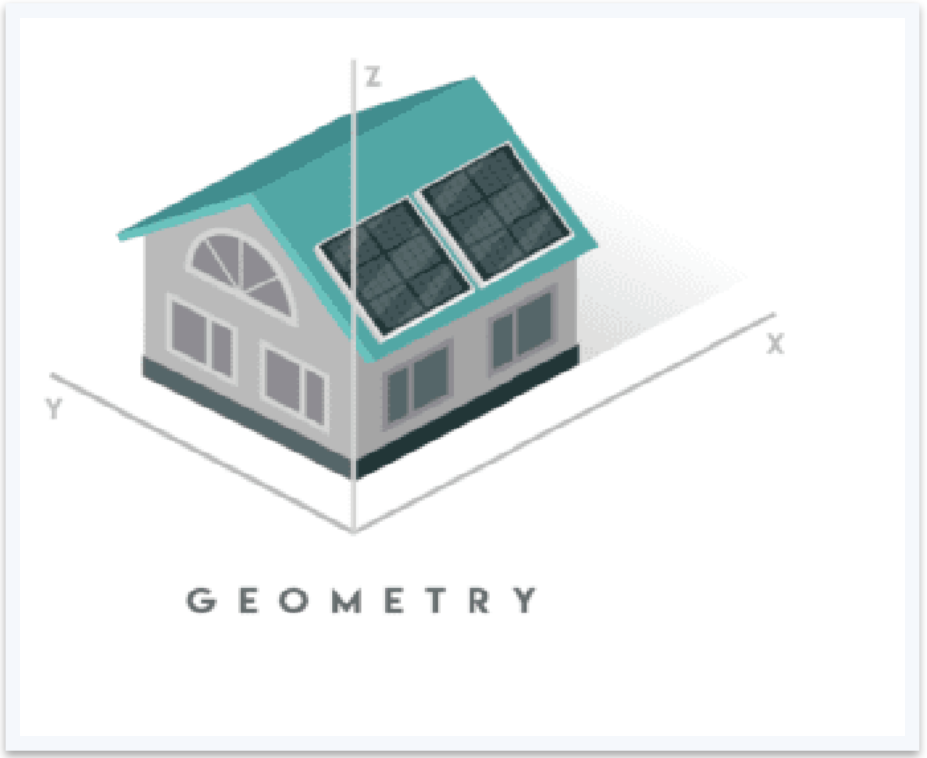


The BIM principles: one shared, coordinated, data-rich model across the project lifecycle.

© 2026 Tertiary Infotech Academy Pte Ltd

BIM Dimensions — 3D Geometry

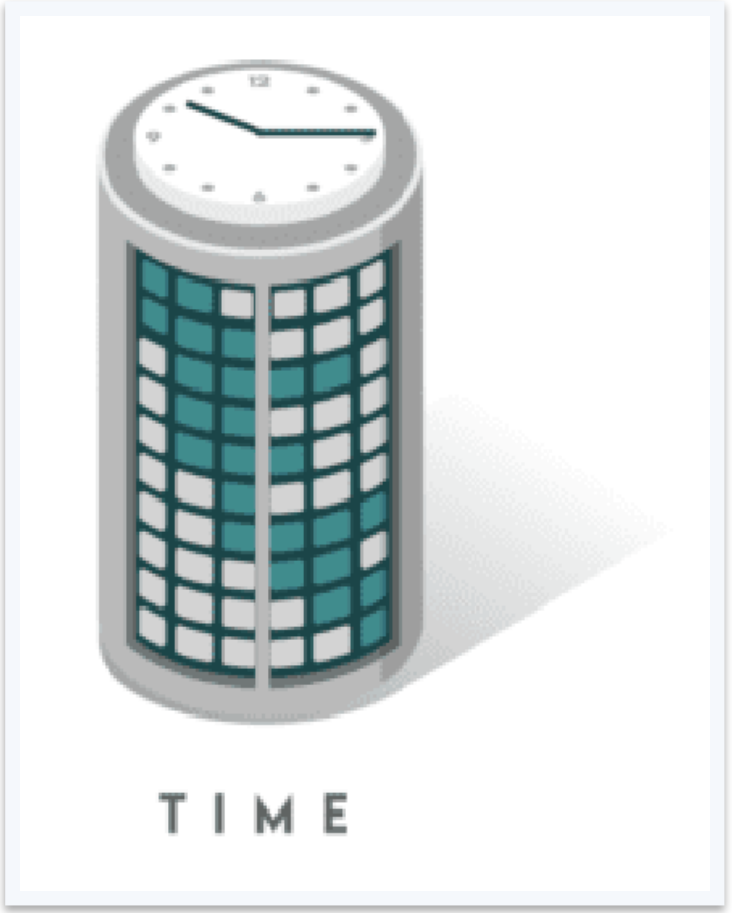
- 3D BIM is the most common model type — the three geographic dimensions (x, y, z) of a structure.
- Helps stakeholders visualise the structure before the project starts.
- Gives AEC professionals the insight and tools to plan, design, construct and manage more efficiently.



© 2026 Tertiary Infotech Academy Pte Ltd

BIM Dimensions — 4D Time

- The 4th dimension adds project duration — timeline and scheduling information show how the project evolves over time.
- Extremely helpful for early-stage conflict detection, site planning and scheduling optimisation.
- Better communicates the next steps to stakeholders during every stage of construction.



© 2026 Tertiary Infotech Academy Pte Ltd

BIM Dimensions — 5D Cost

- 5D BIM visualises budget analysis and cost estimation — analysing project costs over time.
- Especially useful early on, to estimate costs of different design or construction scenarios.
- Tracks predicted vs actual cost with real-time notifications of cost overrun.



© 2026 Tertiary Infotech Academy Pte Ltd

BIM Dimensions — 6D Sustainability

- 6D BIM makes a structure self-sustainable and energy-efficient.
- Stakeholders analyse energy consumption and estimate energy usage during the initial design phase.
- Reduces energy consumption in the long run and improves operational management after construction.



© 2026 Tertiary Infotech Academy Pte Ltd

BIM Dimensions — 7D Facility Management

- 7D BIM improves operations and facility management for building owners and managers.
- Tracks asset data: status, warranty information, technical specifications, maintenance and operation manuals.
- Improves the maintenance process by listing all replacement parts and specifications.



© 2026 Tertiary Infotech Academy Pte Ltd

BIM Dimensions — 8D Safety, and the Full Stack

1

3D — Geometry

The shared x, y, z model every stakeholder visualises.

2

4D — Time

Schedule linked to the model; the build sequence becomes visible.

3

5D — Money

Cost estimation and budget tracking over time.

4

6D — Sustainability

Energy analysis from the design stage onwards.

5

7D — Facility Management

Asset data for operations and maintenance.

6

8D — Safety

Safety information added at design and execution — an overall picture of site risks and hazards for workers, before they occur.

© 2026 Tertiary Infotech Academy Pte Ltd

Value Proposition — Advantages of BIM (1–4)

1

Maximised efficiency

Shorter project lifecycles; pre-construction and planning phases become easier to manage — everyone works off one up-to-date model, reducing errors and rework.

2

Reduced costs and wastage

Better material choices and streamlined construction work minimise human error and wasted materials — cutting cost.

3

Improved cost estimates

Model-based estimates from a detailed 3D model are more realistic and accurate, and quantity takeoffs become far easier.

4

Better insight into the project

A realistic 3D look at the eventual outcome helps contractors and clients feel the built asset — changes happen in pre-construction, not on site.

© 2026 Tertiary Infotech Academy Pte Ltd

VALUE PROPOSITION · K2

Value Proposition — Advantages of BIM (5–7)

1

Communication & collaboration

Cloud-based BIM lets everyone access up-to-date models anywhere, any time — estimates, models and design notes in one place.

2

Less risk

Closer collaboration reduces tender risk premiums, improves on-site safety and removes the risk of building from outdated information.

3

Better end results

Better planning and earlier visualisation produce a higher-quality build and a higher-quality built asset.

4

The common thread

Every advantage flows from one source: a single shared intelligent model that all stakeholders trust.

© 2026 Tertiary Infotech Academy Pte Ltd

The BIM Execution Plan (BEP)

- The BEP is the baseline document, approved by the Employer, that guides the project team in achieving the goals set — including BIM deliverables — throughout the project.
- It specifies the roles and responsibilities of project members when using BIM at different stages.
- It details how deliverables are created, coordinated, maintained and shared to satisfy the project goals.
- Defined at project start; updated (with the Employer's or BIM Manager's permission) when members or BIM uses change.



© 2026 Tertiary Infotech Academy Pte Ltd

REQUIREMENTS · K3

Typical Content of a BEP

1. Project information
2. Project members
3. Project goals & BIM uses for each project stage
4. BIM deliverables for each BIM use
5. Model author and users for each deliverable
6. Model elements, level of detail and attributes
7. Process for BIM creation, maintenance and collaboration
8. Exchange protocol, submittal format, technology infrastructure & software

© 2026 Tertiary Infotech Academy Pte Ltd

REQUIREMENTS · K3

Requirements of BIM — the Information Chain

1

OIR — Organisational Information Requirements

What the organisation needs to know to run its business.

2

AIR — Asset Information Requirements

What the organisation needs to know about the assets it is responsible for.

3

EIR — Employer Information Requirements

The information to be delivered, and the standards and processes to be adopted, for a construction project.

4

BEP — BIM Execution Plan

The overall vision and implementation details the project team follows throughout the project.

5

PIM — Project Information Model

The information model developed during the design and construction phase.

6

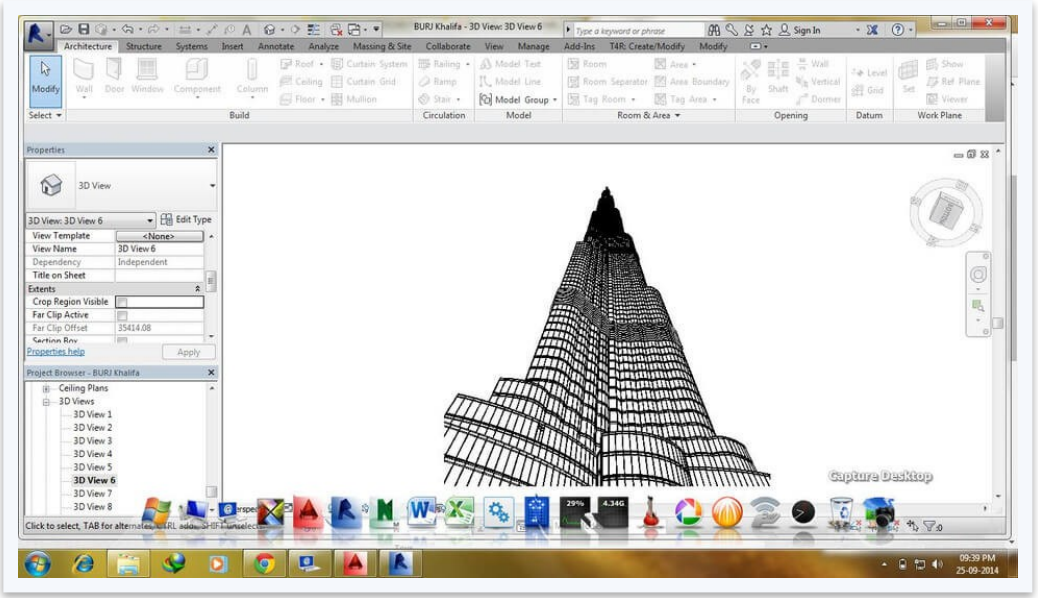
AIM — Asset Information Model

Graphical and non-graphical data, documents and metadata that support asset management.

© 2026 Tertiary Infotech Academy Pte Ltd

What is Revit?

- A BIM-based platform used by architects, engineers, contractors and designers to create one unified model of real-life information.
- A design and documentation platform: one integrated building information model to conceptualise, design and document a project.
- Virtual modelling — see the building or infrastructure before it is built on-site.
- Its tools are built for BIM: an intelligent model with data stored in it — helping you avoid rework, minimise cost, detect conflicts before construction and avoid delays.



© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL DO

Hands-On Labs — Introduction to BIM and Revit

Lab 1

- Get Started with Revit

Lab 2

- Convert 2D Information into a 3D Model

—

- —

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 01 · HANDS-ON

Get Started with Revit

LAB 1

Tour the Revit environment on a real model: open the Autodesk sample architecture project, learn the ribbon, Properties palette, Project Browser and drawing area, then navigate plans, elevations, sections and the 3D view.

You'll build

A guided tour of a complete BIM model — you can open any view, orbit the 3D model and inspect element properties.

Tools: Autodesk Revit 2024/2025, Autodesk sample project (rac_basic_sample_project.rvt)

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 1 · STEP-BY-STEP

Get Started with Revit

1

of 6

Launch Revit. On the Home screen under Models click Open, browse to the Revit sample files and open the basic architecture sample project.

Home ▸ Models ▸ Open ▸ rac_basic_sample_project.rvt

2

of 6

Identify the four working zones: the ribbon (tools), the Properties palette (instance/type properties), the Project Browser (all views, schedules, sheets, families) and the drawing area.

© 2026 Tertiary Infotech Academy Pte Ltd

Get Started with Revit

3

of 6

In the Project Browser expand Floor Plans and double-click Level 1. Zoom and pan with the mouse wheel; hover over a wall and read its description on the status bar.

Project Browser ▸ Floor Plans ▸ Level 1

4

of 6

Select a wall and inspect the Properties palette: note its Type (in the Type Selector), Base Constraint and Top Constraint. Click Edit Type to view (not change) the type parameters.

Properties ▸ Type Selector / Edit Type

© 2026 Tertiary Infotech Academy Pte Ltd

Get Started with Revit

5

of 6

Open the default 3D view and orbit the model with Shift + middle-mouse drag. Use the ViewCube to jump to Front, Top and corner views.

Quick Access Toolbar ▸ Default 3D View

6

of 6

Open an elevation and a section view from the Project Browser, and tile the windows to see the same model in several views at once.

View ▸ Windows ▸ Tile Views

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 1 · VERIFY

Get Started with Revit

✅ Test it

You can name the four zones of the Revit UI, open any plan/elevation/section/3D view from the Project Browser, and read a selected element's type and constraints in the Properties palette — one model driving every view is the essence of BIM.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 01 · HANDS-ON

Convert 2D Information into a 3D Model

LAB 2

Turn a 2D PDF floor plan into an intelligent 3D model: create gridlines and walls to match the drawing, set the exterior walls to the 12" Generic type and interior walls to the 6" Generic type, constrain wall tops to Level 2 and give the corner a 30 ft radius — then add construction information to the model.

You'll build

A 3D building model reproduced from a 2D plan, with correctly typed exterior/interior walls constrained to levels.

Tools: Autodesk Revit, 2D floor plan PDF (from the LMS lab files)

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 2 · STEP-BY-STEP

Convert 2D Information into a 3D Model

1

of 6

Start a new project from the Architectural template and link the 2D floor plan PDF into the Level 1 plan view as a reference underlay.

File ▸ New ▸ Project ▸ Architectural Template · Insert ▸ Link PDF

2

of 6

Place vertical and horizontal gridlines to match the grid intersections shown on the PDF.

Architecture ▸ Datum ▸ Grid

© 2026 Tertiary Infotech Academy Pte Ltd

Convert 2D Information into a 3D Model

3

of 6

Draw the exterior walls over the PDF using the 12" Generic wall type. In the Options Bar set Height to Level 2.

Architecture ▸ Build ▸ Wall: Architectural ▸ Generic - 12"

4

of 6

Draw the interior partition walls with the 6" Generic wall type, also constrained to Level 2.

Architecture ▸ Build ▸ Wall: Architectural ▸ Generic - 6"

© 2026 Tertiary Infotech Academy Pte Ltd

Convert 2D Information into a 3D Model

5

of 6

Create the curved corner wall with the Fillet Arc draw tool and set the radius to 30 ft (30' 0").

Modify | Place Wall ▸ Draw ▸ Fillet Arc ▸ Radius = 30'

6

of 6

Open the default 3D view and verify the model: all walls rise from Level 1 to Level 2 and the plan matches the PDF. Select any wall and confirm its Top Constraint reads 'Up to level: Level 2'.

Quick Access Toolbar ▸ Default 3D View

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 2 · VERIFY

Convert 2D Information into a 3D Model

✅ Test it

The 3D view shows a closed building shell that matches the 2D PDF: exterior walls are Generic - 12", interior walls are Generic - 6", every wall's Top Constraint is Level 2, and the corner wall has a 30 ft radius — 2D information has become an intelligent, parametric 3D model.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC RECAP

Recap — Introduction to BIM and Revit

- You can now: LO1 — Apply Revit for Building Information Modelling (BIM)

© 2026 Tertiary Infotech Academy Pte Ltd

02

TOPIC 02

BIM Modeling

Develop models for building elements · walls, floors, ceilings, roofs, stairs, structure · site and topography · massing · BIM design integration

© 2026 Tertiary Infotech Academy Pte Ltd

ALIGNMENT: K6, K7 · A4, A5 · LO2

Key Concepts — BIM Modeling

1

Everything is parametric

Each component has parametric qualities — change the model once and every related view, schedule and sheet updates automatically.

2

Model with the right tool

Walls, curtain walls, floors, ceilings, roofs, stairs, railings — each element is created with its dedicated tool and family type.

3

Structure lives in the model

Structural columns, beams, beam systems, trusses and braces carry the load path inside the same coordinated model.

4

Site gives context

Toposurfaces, subregions, graded regions and building pads model the terrain the building sits on.

5

Mass first, detail later

Conceptual masses and generative design (Dynamo) explore building volumes before committing to detailed elements.

6

Families drive integration

System, loadable and in-place families — with materials and parameters — integrate design information into every element.

© 2026 Tertiary Infotech Academy Pte Ltd

Develop Models for Building Elements

1

Parametric objects

Each component has parametric qualities that drive its behaviour.

2

Quick changes, no repetition

One change updates every related component, view, detail and document automatically.

3

Construction documentation

High-quality documentation produced automatically from the 3D model.

4

Quantities & cost estimation

Generated automatically from the data inserted in the model.

5

Coordination & 3D visualisation

Shared models reduce conflicts and communicate design to client and team.

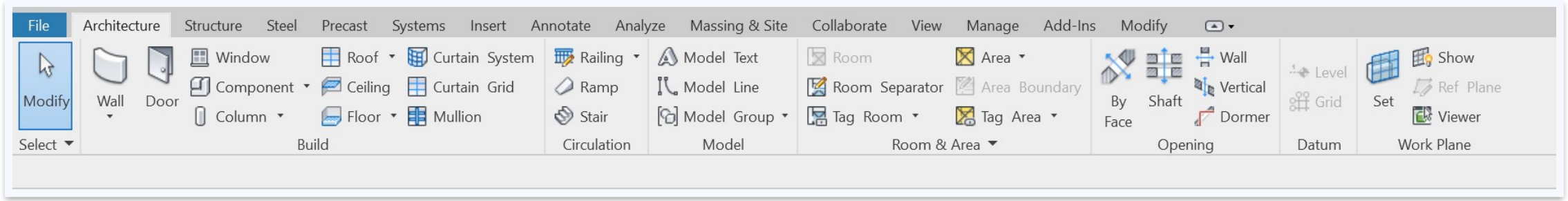
6

Interoperability

Revit imports, exports and links data in common formats such as IFC.

© 2026 Tertiary Infotech Academy Pte Ltd

Modeling Walls

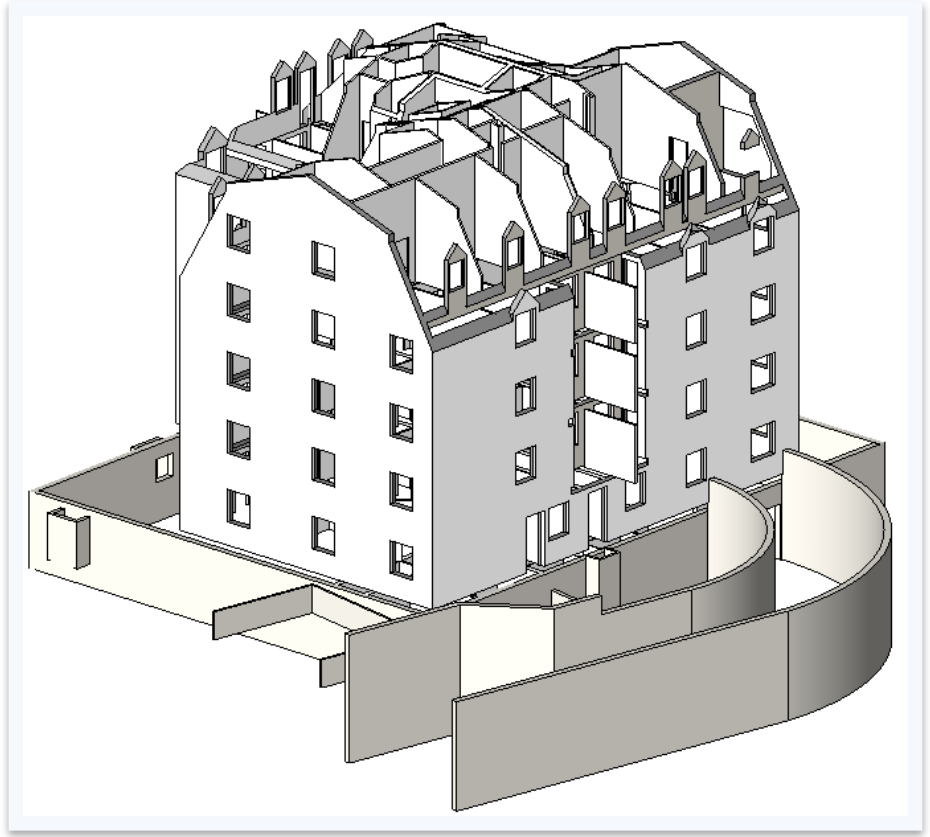


- Add instances of a wall type in a floor plan or 3D view: Architecture ▸ Build ▸ Wall ▸ Wall: Architectural (structural walls work the same way).
- Pick the wall type in the Type Selector; adjust instance properties in the Properties palette before placing.
- Options Bar: Level (base constraint), Height (top constraint), Location Line (which plane follows the cursor), Chain, Offset and Join Status.

© 2026 Tertiary Infotech Academy Pte Ltd

Wall Placement Options

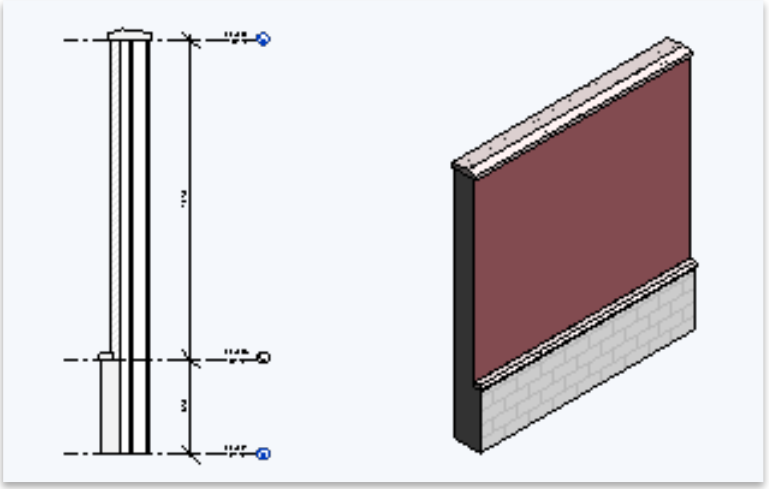
- Location Line — align the wall's centreline, core face or finish face with the cursor as you draw.
- Chain — draw a series of wall segments connected at endpoints.
- Offset — place the wall's location line at a distance from the cursor or a picked line/face.
- Join Status — Allow creates a butt join where walls intersect; Disallow prevents joining.



© 2026 Tertiary Infotech Academy Pte Ltd

Stacked and Embedded Walls

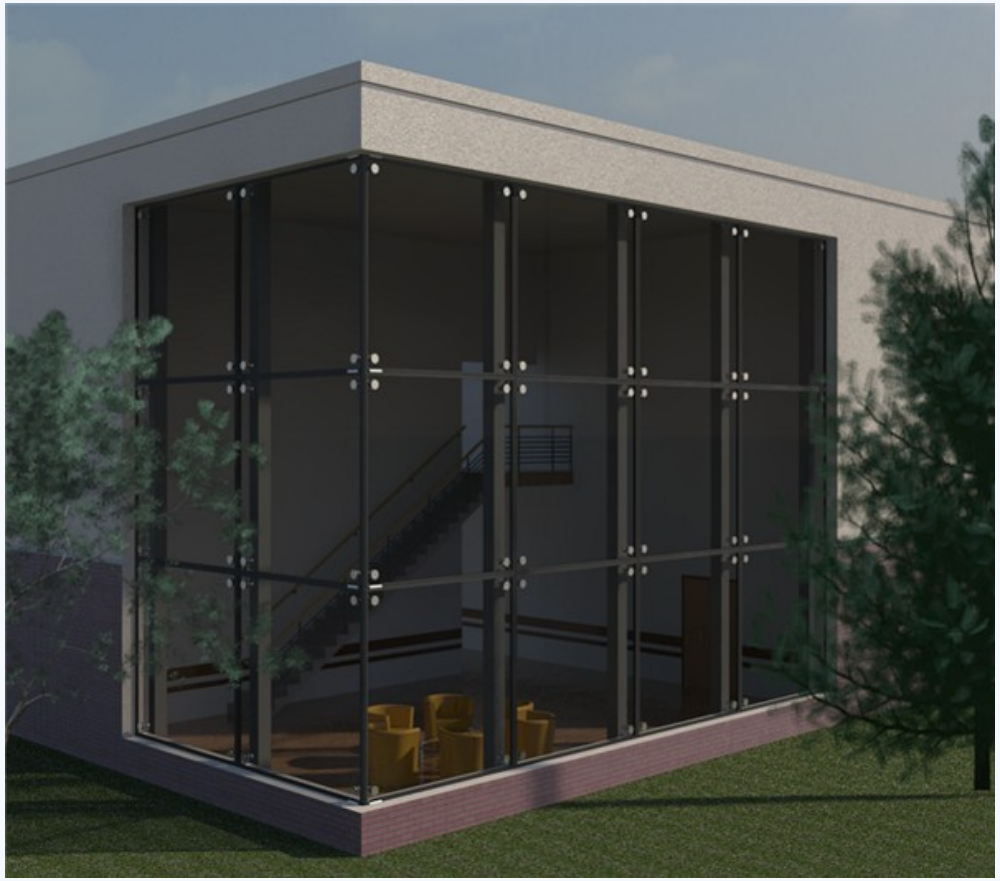
- Stacked walls: two or more subwalls of different thicknesses stacked at different heights — attached, with joined geometry.
- Embed a wall into a host wall (wall-in-wall, wall-in-curtain-wall or curtain-wall-in-wall).
- The embedded wall must be smaller than its host; use Cut Geometry, picking the HOST wall first, then the wall to embed.
- Resize an embedded wall with its drag controls after embedding.



© 2026 Tertiary Infotech Academy Pte Ltd

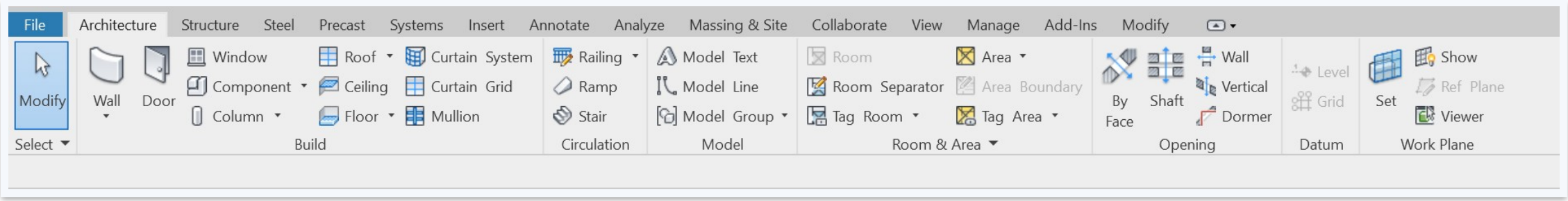
Modeling Curtain Walls

- A curtain wall is placed like any wall — just select a curtain wall type (Curtain Wall 1, Exterior Glazing, Storefront) in the Type Selector.
- Divide the wall into panels with Curtain Grid; customise grids with Add/Remove Segments.
- Place mullions on grid lines; Tab-select any panel to change its type — even to a door panel.
- Doors and walls can be embedded within curtain panels.



© 2026 Tertiary Infotech Academy Pte Ltd

Modeling Floors and Openings

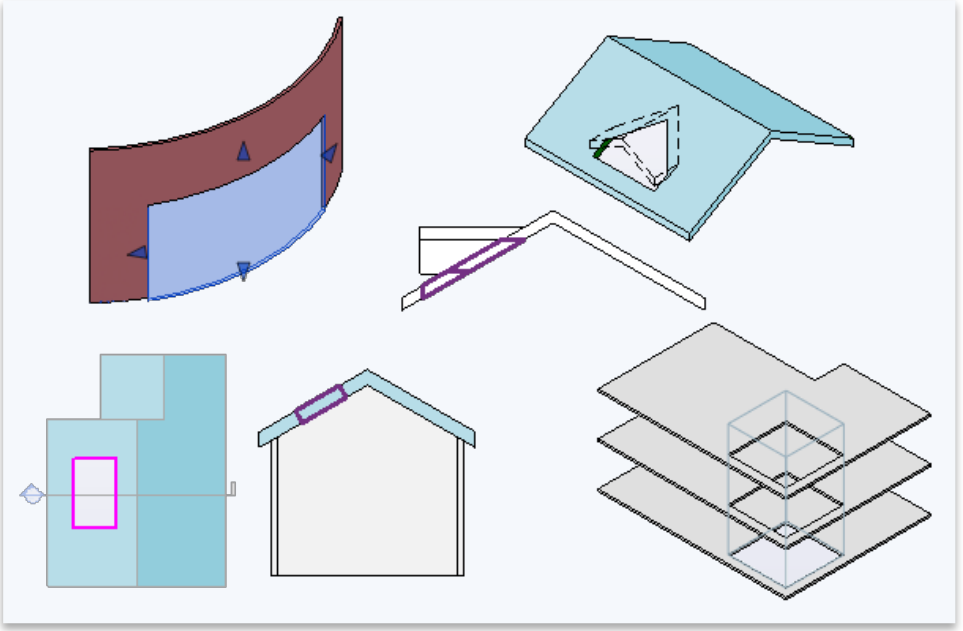


- Create a floor by picking walls or sketching its boundary — the boundary must be a closed loop; sketch a second loop inside to form an opening.
- Options Bar: Offset for floor edges; 'Extend into wall (to core)' measures the offset from the wall's core.
- Opening tools: By Face (perpendicular to the face), Vertical (perpendicular to a level), Shaft (cuts every level it spans), Wall Opening and Dormer.

© 2026 Tertiary Infotech Academy Pte Ltd

Shaft Openings and View Range

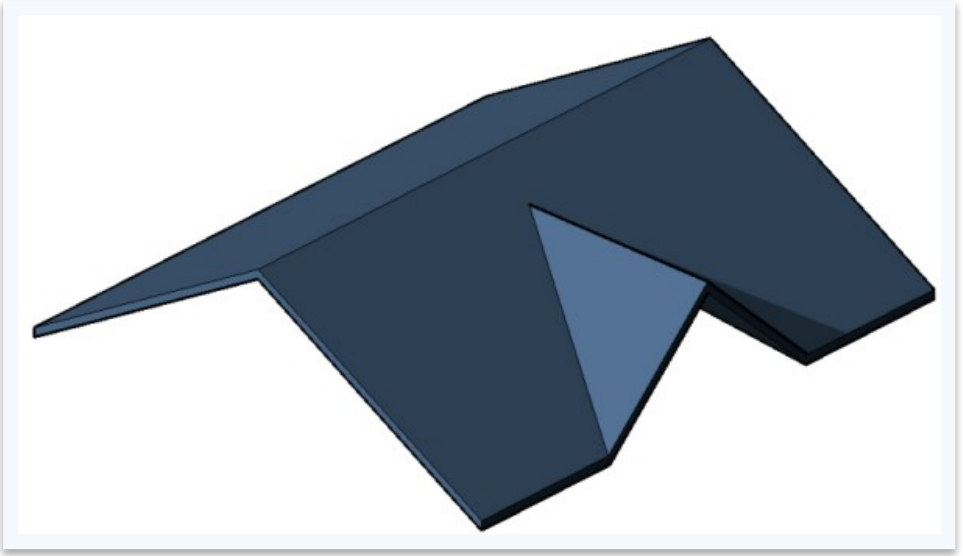
- A Shaft cuts through roofs, floors and ceilings across its Base and Top Constraints — move it on one level and it moves on all levels.
- Every plan view has a View Range: Top, Cut Plane and Bottom planes define what is shown and how.
- View Depth adds a plane beyond the primary range to show elements below the bottom clip plane.



© 2026 Tertiary Infotech Academy Pte Ltd

Modeling Ceilings and Roofs

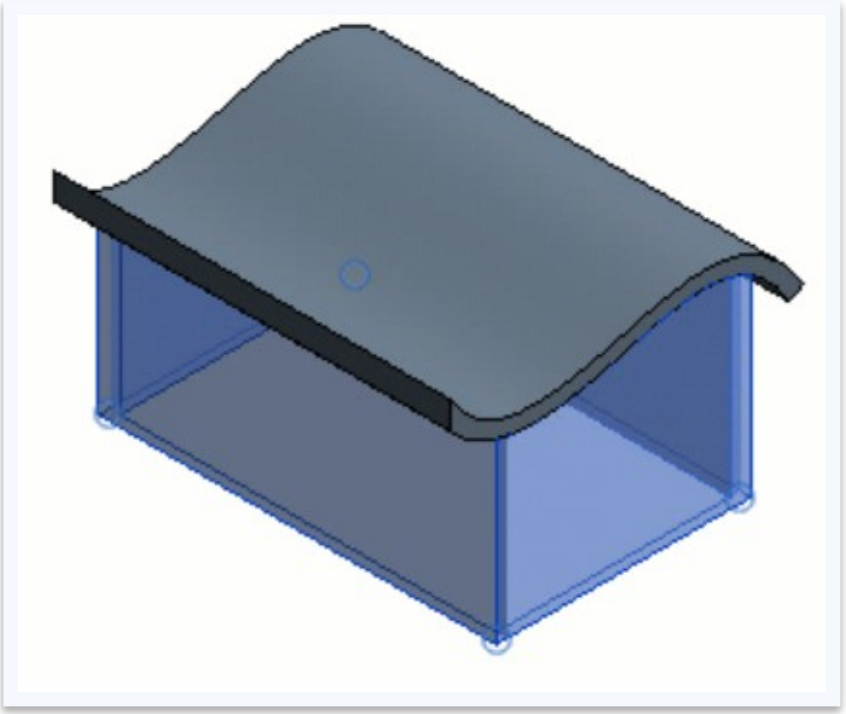
- Ceilings are placed in a reflected ceiling plan — Automatic Ceiling fills a closed wall loop; Sketch Ceiling draws a custom boundary.
- Roof by Footprint: sketch/pick a closed loop in a plan view; slope-defining edges create hips and gables.
- A roof cannot cut through windows or doors.



© 2026 Tertiary Infotech Academy Pte Ltd

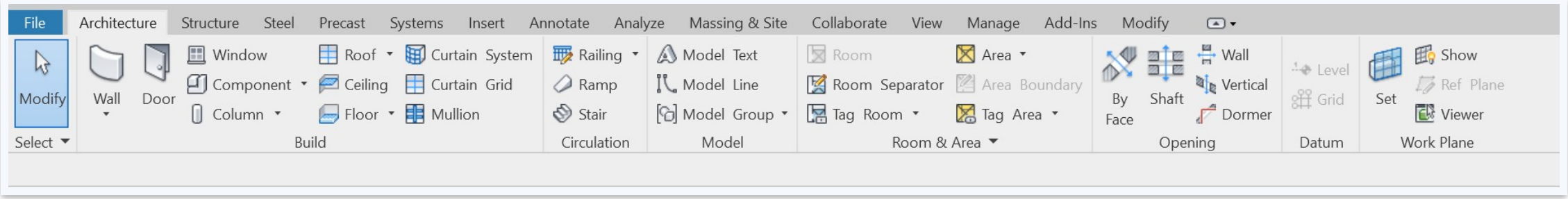
Roof by Extrusion

- Create a roof by extruding an open-loop profile sketched in an elevation, 3D or section view.
- Set the Roof Reference Level and Offset — a reference plane controls the roof's position.
- After creating it, attach walls to the roof, rehost it or edit its work plane.



© 2026 Tertiary Infotech Academy Pte Ltd

Modeling Stairs and Railings



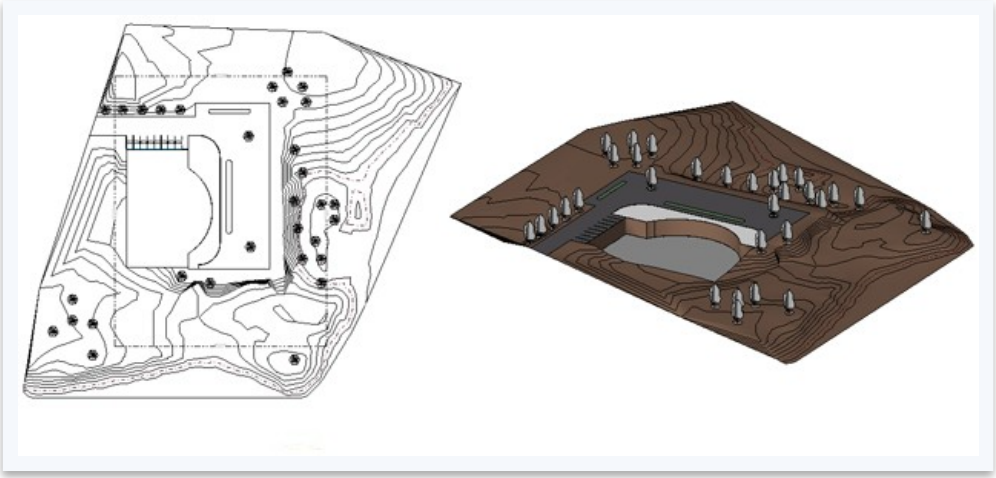
- Stairs assemble from components in edit mode: runs (straight, spiral, U-, L-shaped, sketched), landings, supports and railings.
- Railings: free-standing on levels, attached to hosts (floors, ramps, stairs), auto-created with stairs, or sketched with a custom path and Pick New Host.
- Tile plan + 3D views while assembling to see the full stair model as you work.

© 2026 Tertiary Infotech Academy Pte Ltd

SITE · A4

Site and Topography

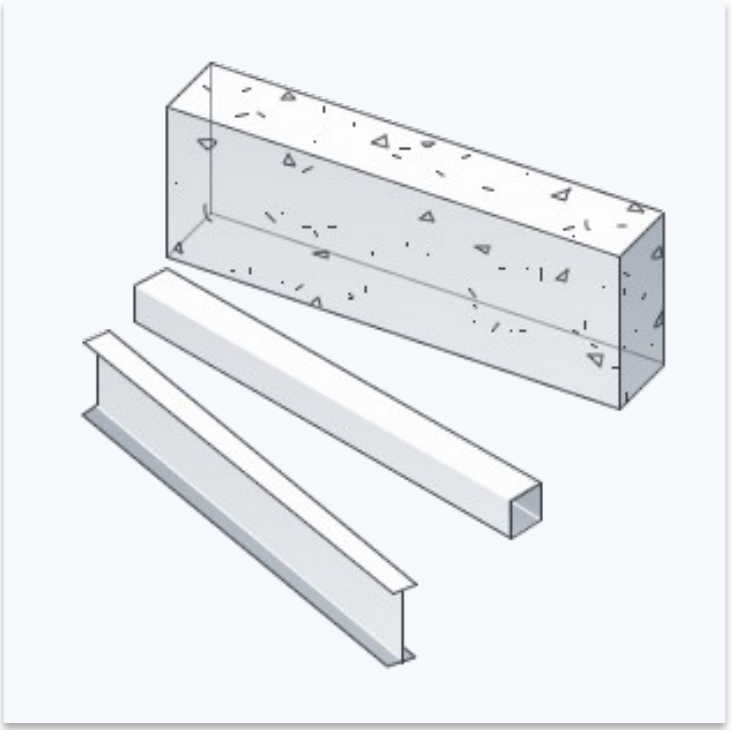
- The Toposurface tool defines terrain from points placed at absolute or relative elevations, in 3D or site plan views.
- Subregion applies different properties (e.g. material) to an area; Split and Merge Surfaces divide and rejoin terrain; Graded Region develops the proposed site design.
- A Building Pad sketched on the toposurface cuts the terrain at a controlled level offset — with openings and slope if needed.



© 2026 Tertiary Infotech Academy Pte Ltd

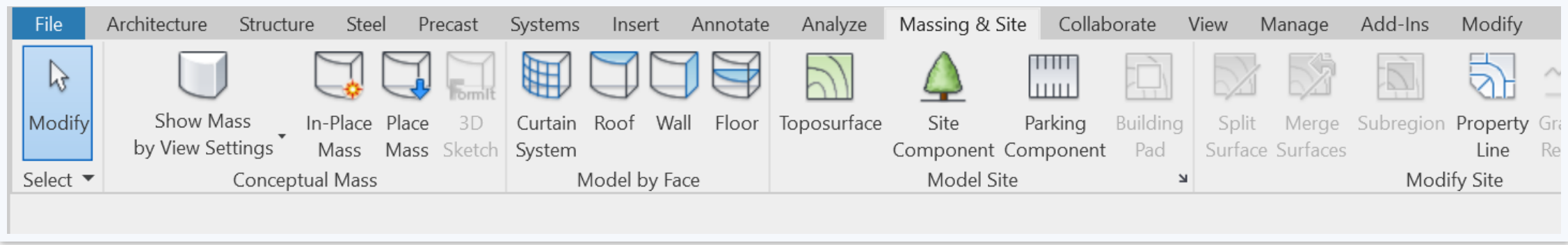
Structural Columns and Framing

- Structural columns add vertical load-bearing elements: Structure ▸ Column, or At Grids for batch placement.
- Beams span supports; a Beam System fills a boundary with members per a layout rule; 3D option for sloped systems.
- Trusses place framing per the selected truss family's layout; braces sketch diagonally between structural elements in plan or framing elevations.



© 2026 Tertiary Infotech Academy Pte Ltd

Conceptual Masses



- Mass modeling drafts conceptual volumes fast — as Conceptual Mass families (outside the project) or In-Place Masses (inside it).
- Use massing studies for design options, phase representation, zoning compliance (visually and numerically) and assembling complex forms from mass families.
- Generate floors, roofs, curtain systems and walls from mass instances — regenerating under control when the mass changes; schedule gross volume, floor area and surface area.

© 2026 Tertiary Infotech Academy Pte Ltd

Generative Design and Dynamo

- Generative design (AEC Collection) generates design alternatives from your inputs, constraints and goals.
- Dynamo is Revit's visual-programming add-in — open-source graphical programming for custom computational design and BIM automation.
- Workflow: build a Dynamo graph, declare Is Input / Is Output on nodes, export for generative design, then review the study's outcomes.



© 2026 Tertiary Infotech Academy Pte Ltd

Families and Parameters

The three kinds

- System families
  - Walls, roofs, floors, ducts, pipes — predefined in Revit, never loaded from external files
  - Levels, grids, sheets and viewports are system families too
- In-place families
  - Unique, project-specific elements created with Family Editor tools
  - Reference other geometry and adjust when it changes

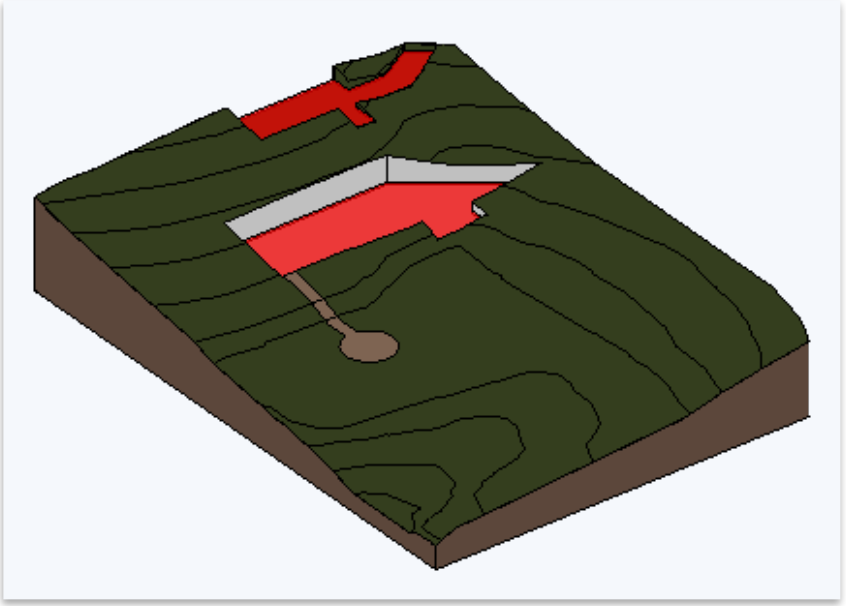
Loading & catalogs

- Loadable families
  - Windows, doors, casework, fixtures, furniture, boilers, plumbing — RFA files loaded via Insert ▸ Load Family
  - Combine into nested and shared families
- Type catalogs
  - Load only the types you need from large families (e.g. one C-Channel size, not dozens)
  - Keeps project size down and the Type Selector short

© 2026 Tertiary Infotech Academy Pte Ltd

Materials

- Materials control how elements display in views and rendered images: Manage ▸ Settings ▸ Materials.
- They define Graphics, Appearance, Thermal and Physical information — for visualisation, analysis and scheduling.
- Apply by category or subcategory, by family, by element parameter, or by face with the Paint tool.



© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL DO

Hands-On Labs — BIM Modeling

Labs 3–4

- Modeling Walls — Reveals, Embedded and Curtain Walls
- Floors, Ceilings, Roofs and Openings

Labs 5–6

- Stairs, Railings and Structural Elements
- Terrain, Building Pad and Conceptual Mass

—

- —

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 02 · HANDS-ON

Modeling Walls — Reveals, Embedded and Curtain Walls

LAB 3

Work with advanced wall modeling on the Embed Walls project: place brick-course wall reveals on an elevation, duplicate a wall type into a new Vertical Cladding type, embed the new walls into an existing wall with Cut Geometry, and add a curtain wall with grids and mullions.

You'll build

A facade with three positioned reveals, embedded Vertical Cladding walls joined into the host wall, and a gridded curtain wall.

Tools: Autodesk Revit, Embed Walls.rvt (LMS lab files)

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 3 · STEP-BY-STEP

Modeling Walls — Reveals, Embedded and Curtain Walls

1

of 6

Open Embed Walls.rvt. Start the Wall: Reveal tool, edit its type properties and change the profile to Reveal-Brick Course: 3 Bricks.

Architecture ▸ Build ▸ Wall ▸ Wall: Reveal ▸ Edit Type

2

of 6

Place three reveals on the end elevation wall at the dimensions given in the drawing, each with Offset from Wall = 1".

Modify | Place Reveal ▸ set Offset from Wall = 1"

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 3 · STEP-BY-STEP

Modeling Walls — Reveals, Embedded and Curtain Walls

3

of 6

Open the Floor 2 plan. Start a new wall, select the Stone on CMU type, duplicate it and name the new type Vertical Cladding; change Layer 1's material to Cladding, Vertical Ribbed.

Architecture ▸ Wall ▸ Edit Type ▸ Duplicate ▸ 'Vertical Cladding'

4

of 6

Draw three lengths of the Vertical Cladding wall per the plan dimensions, with Base Constraint = Floor 2 and Top Constraint = Roof, then Align their tops/bottoms to the reveals.

Modify ▸ Align

© 2026 Tertiary Infotech Academy Pte Ltd

Modeling Walls — Reveals, Embedded and Curtain Walls

5

of 6

Use Cut Geometry to embed the Vertical Cladding walls into the existing host wall (pick the host first), then Join Geometry to remove the end edges in plan.

Modify ▸ Geometry ▸ Cut / Join

6

of 6

Add a curtain wall: place a wall with a curtain wall type, divide it with Curtain Grid, and place mullions on the grid lines. Tab-select a panel and swap its type in the Type Selector.

Architecture ▸ Wall ▸ Curtain Wall 1 · Architecture ▸ Build ▸ Curtain Grid / Mullion

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 3 · VERIFY

Modeling Walls — Reveals, Embedded and Curtain Walls

✅ Test it

The elevation shows three brick-course reveals at the specified dimensions; the Vertical Cladding walls sit flush inside the host wall with clean joins in plan; and the curtain wall shows a grid of panels with mullions — Tab-selection lets you re-type any single panel.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 02 · HANDS-ON

Floors, Ceilings, Roofs and Openings

LAB 4

Complete the horizontal envelope of the building: create a floor by picking walls, add a ceiling in a reflected ceiling plan, create a roof by footprint with slope-defining edges and a second roof by extrusion, then cut a vertical shaft opening through the levels.

You'll build

A building with floors, ceilings, a footprint roof and an extruded roof, plus a shaft opening cutting through every level.

Tools: Autodesk Revit, your Lab 2 model (or the LMS lab file)

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 4 · STEP-BY-STEP

Floors, Ceilings, Roofs and Openings

1

of 6

Create the Level 1 floor: pick the exterior walls as boundaries (Extend into wall (to core)), and finish the sketch — the boundary must be a closed loop.

Architecture ▸ Build ▸ Floor: Architectural ▸ Pick Walls

2

of 6

Open the Level 1 ceiling plan and place a ceiling with Automatic Ceiling by clicking inside the walls that form a closed loop.

Architecture ▸ Build ▸ Ceiling ▸ Automatic Ceiling

© 2026 Tertiary Infotech Academy Pte Ltd

Floors, Ceilings, Roofs and Openings

3

of 6

Create the main roof by footprint on Level 2: pick the walls with an overhang, keep Defines Slope on for all edges, and finish — then open 3D to see the hips form.

Architecture ▸ Build ▸ Roof ▸ Roof by Footprint

4

of 6

Create a porch roof by extrusion: in an elevation view sketch an open-loop profile and extrude it, then Attach the porch walls to the underside of the roof.

Architecture ▸ Build ▸ Roof ▸ Roof by Extrusion · Modify Wall ▸ Attach Top/Base

© 2026 Tertiary Infotech Academy Pte Ltd

Floors, Ceilings, Roofs and Openings

5

of 6

Cut a shaft for the stair: sketch a closed loop with the Shaft tool, set Base Constraint = Level 1 and Top Constraint = Roof, and finish the opening.

Architecture ▸ Opening ▸ Shaft

6

of 6

Verify in a section view that the shaft cuts the floor and ceiling on every intermediate level, and use By Face / Vertical openings where a duct or chimney penetrates the roof.

Architecture ▸ Opening ▸ By Face / Vertical

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 4 · VERIFY

Floors, Ceilings, Roofs and Openings

✅ Test it

The section view shows floor, ceiling and both roofs in place; the shaft opening cuts cleanly through every level it spans; and moving the shaft in one plan moves it on all levels — the model stays coordinated because every view reads the same database.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 02 · HANDS-ON

Stairs, Railings and Structural Elements

LAB 5

Add vertical circulation and the load path: assemble a stair from runs and landings, host a railing on a floor edge with Sketch Path, then place structural columns on the grid intersections and frame them with beams, a beam system and a brace.

You'll build

A stair with railings connecting two levels, plus a structural frame of columns, beams, a beam system and a brace.

Tools: Autodesk Revit, your Lab 2/4 model (or the LMS lab file)

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 5 · STEP-BY-STEP

Stairs, Railings and Structural Elements

1

of 6

Create a stair between Level 1 and Level 2: in stair assembly mode place a straight Run; a landing is created automatically between two runs. Finish and inspect it in 3D.

Architecture ▸ Circulation ▸ Stair ▸ Run

2

of 6

Add a railing along the Level 2 floor edge: Sketch Path, draw the railing line, use Pick New Host to host it on the floor, and finish.

Architecture ▸ Circulation ▸ Railing ▸ Sketch Path ▸ Pick New Host

© 2026 Tertiary Infotech Academy Pte Ltd

Stairs, Railings and Structural Elements

3

of 6

Place structural columns at grid intersections from Level 1 to Level 2 (use At Grids to place several at once).

Structure ▸ Structure ▸ Column ▸ At Grids

4

of 6

Draw beams between column tops on Level 2, snapping from column to column.

Structure ▸ Structure ▸ Beam

© 2026 Tertiary Infotech Academy Pte Ltd

Stairs, Railings and Structural Elements

5

of 6

Fill a bay with a Beam System: sketch its boundary with Pick Supports, set the Beam Type and Layout Rule (e.g. Fixed Distance), and finish.

Structure ▸ Structure ▸ Beam System ▸ Sketch Beam System

6

of 6

Open a framing elevation and add a diagonal Brace between a column and a beam, snapping to their ends.

Structure ▸ Structure ▸ Brace

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 5 · VERIFY

Stairs, Railings and Structural Elements

✅ Test it

The stair connects the two levels with automatically generated railings; columns stand on every chosen grid intersection; the beam system spaces its members per the layout rule; and the brace runs diagonally in the framing elevation — the architectural and structural elements coexist in one coordinated model.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 02 · HANDS-ON

Terrain, Building Pad and Conceptual Mass

LAB 6

Set the building into its site: create a toposurface by placing points at different elevations, add a building pad from the foundation footprint, frame the 3D view with a section box, apply a grass material — then explore an in-place conceptual mass and preview generative design with Dynamo.

You'll build

A site terrain with a building pad cut into it, a realistic 3D view, and an in-place conceptual mass volume.

Tools: Autodesk Revit, GSG_03_Terrain_Pad.rvt (LMS lab files), Dynamo

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 6 · STEP-BY-STEP

Terrain, Building Pad and Conceptual Mass

1

of 6

Open GSG_03_Terrain_Pad.rvt and the Site plan. Start Toposurface and place points around the building at Elevation 0, then rings of points further out at higher elevations to slope the ground.

Massing & Site ▸ Model Site ▸ Toposurface ▸ Place Point

2

of 6

Finish the surface and open the 3D view — the terrain now buries part of the building.

Massing & Site ▸ Toposurface ▸ ✓ Finish Surface

© 2026 Tertiary Infotech Academy Pte Ltd

Terrain, Building Pad and Conceptual Mass

3

of 6

Add a Building Pad: sketch its boundary with Pick Walls on the foundation walls, set the pad level, and finish — the pad cuts the terrain away from the building.

Massing & Site ▸ Model Site ▸ Building Pad

4

of 6

Frame the model with a Section Box in the 3D view's properties and drag its grips to crop the terrain neatly around the building.

3D View ▸ Properties ▸ Section Box

© 2026 Tertiary Infotech Academy Pte Ltd

Terrain, Building Pad and Conceptual Mass

5

of 6

Select the toposurface, and in the Material parameter choose a grass material for a realistic render, then explore Subregion, Split Surface and Graded Region on the Massing & Site tab.

Toposurface ▸ Properties ▸ Material ▸ Grass

6

of 6

Create an In-Place Mass: name it, sketch a profile and create a form, then Finish Mass — and open Manage ▸ Dynamo to see how a graph of nodes can drive such geometry for generative design.

Massing & Site ▸ Conceptual Mass ▸ In-Place Mass · Manage ▸ Dynamo

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 6 · VERIFY

Terrain, Building Pad and Conceptual Mass

✅ Test it

The 3D view shows the building sitting in a grass-covered terrain with a clean pad cut into the slope, the section box crops the site tidily, and your in-place mass appears as a volume you could later turn into walls, floors and roofs — site, mass and building live in one model.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC RECAP

Recap — BIM Modeling

- You can now: LO2 — Develop models and integrate design for BIM

© 2026 Tertiary Infotech Academy Pte Ltd

03

TOPIC 03

BIM Application

Apply and operate BIM · technologies used in BIM · interpret output · schedules and quantity takeoffs · analyse performance and check compliance

© 2026 Tertiary Infotech Academy Pte Ltd

ALIGNMENT: K5 · A6, A7, A8 · LO3

Key Concepts — BIM Application

1

BIM manages the build

Construction simulation, information statistics and real-time monitoring strengthen quality and process management on site.

2

Views interpret the model

View range, view templates, callouts and section views turn one model into many purpose-built, readable outputs.

3

Schedules read the database

Schedules and quantity takeoffs tabulate elements straight from the model — the model IS the quantity database.

4

Cost from the model

Calculated-value formulas and material takeoffs produce preliminary cost estimates that update as the design changes.

5

Analyse before you build

Revit Systems Analysis translates the model to EnergyPlus to simulate heating, cooling and annual energy performance.

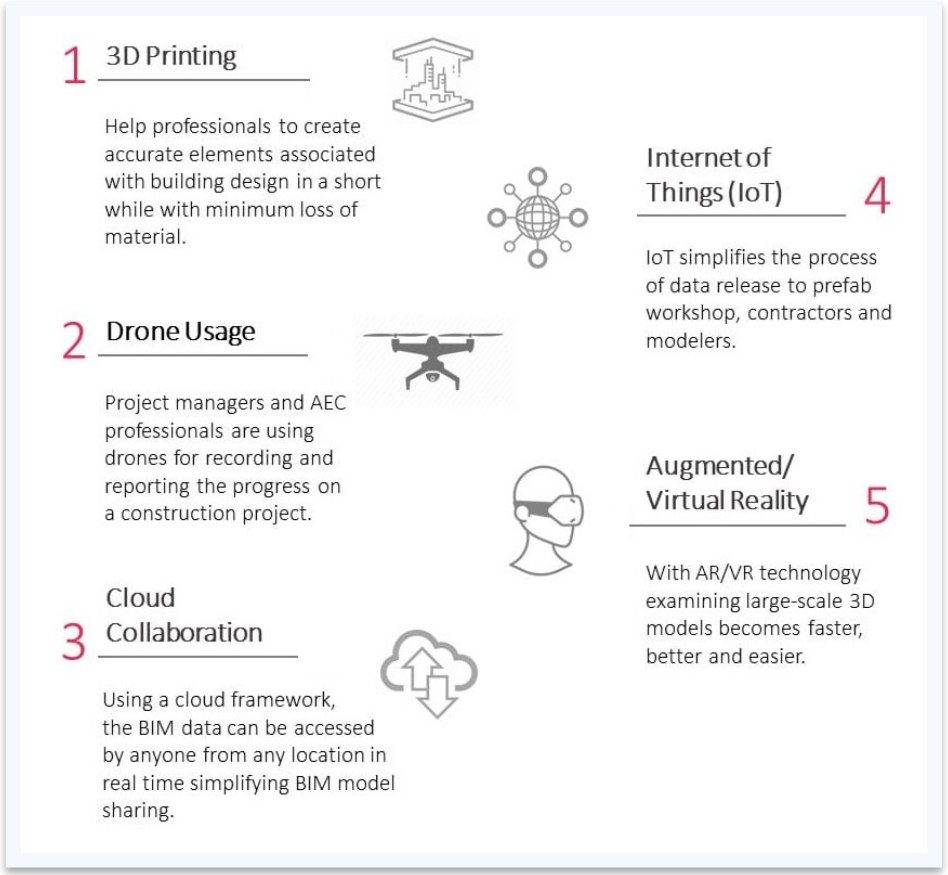
6

Check, then comply

Interference (clash) checks and analysis reports verify design performance and compliance before construction.

© 2026 Tertiary Infotech Academy Pte Ltd

Technologies Used in BIM



The technology stack around BIM: authoring, analysis, collaboration, field and FM tools.

© 2026 Tertiary Infotech Academy Pte Ltd

APPLICATION · K5

Application of BIM in Construction Management

- BIM supports construction simulation and information statistics — management processes become visual and controllable.
- Quality management covers both the product and the technical process.
- Real-time monitoring of the construction process helps ensure construction quality on site.

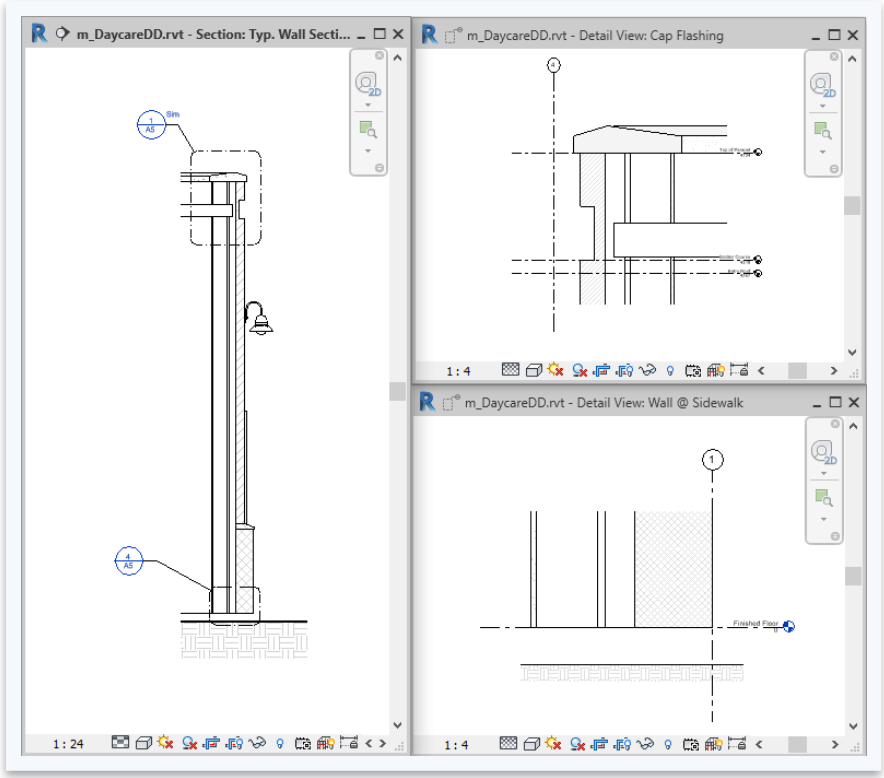


© 2026 Tertiary Infotech Academy Pte Ltd

INTERPRET OUTPUT · A7

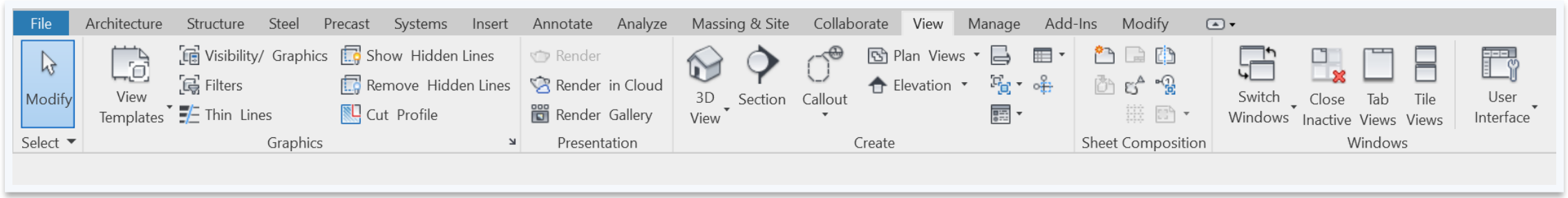
Interpreting the Model — View Range and Callouts

- View Range (Top, Cut Plane, Bottom + View Depth) controls what a plan view shows and how elements are cut.
- Callouts enlarge part of a plan, section, detail or elevation — the callout tag links parent view to callout view.
- Delete the parent view and its callouts go with it; load a callout head to place callout tags.



© 2026 Tertiary Infotech Academy Pte Ltd

Section Views



- Add a section line and crop region to define a new section view: View ▸ Create ▸ Section, drag through the model, resize the crop with the blue controls.
- Open the section by double-clicking its header or from the Project Browser's Sections group.
- The section view updates whenever the design changes or the section line moves.

© 2026 Tertiary Infotech Academy Pte Ltd

Creating a Schedule or Quantity

1. View ▸ Schedules ▸ Schedule/Quantities
2. Pick the category and phase
3. Select fields to report
4. Filter, sort and group the data
5. Format the schedule's appearance
6. The schedule fills itself from the model

© 2026 Tertiary Infotech Academy Pte Ltd

Quantity Takeoffs and Cost Estimates

1

Quantity takeoffs

Essential for accurate construction cost estimating — quantify elements from the model and track element-status changes with ongoing analysis.

2

Material takeoffs

Tabulate materials and areas across categories — more comprehensive than component schedules alone.

3

Calculated values

Formula-driven reporting fields for schedules and tag labels — e.g. Cost = Area × rate; the Name becomes the column header.

4

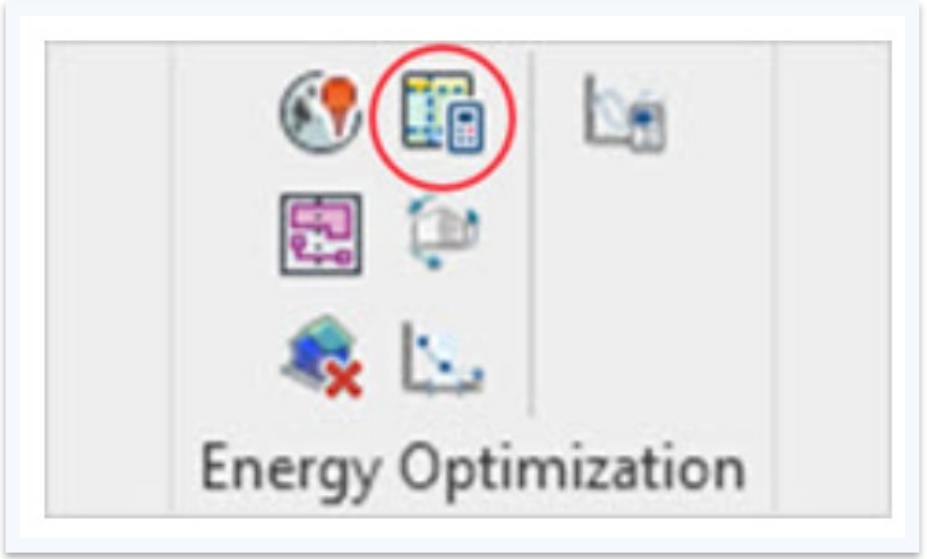
Massing studies

Mass floor schedules report gross volume, floor area and surface area — preliminary cost feedback that updates as the mass changes.

© 2026 Tertiary Infotech Academy Pte Ltd

Revit Systems Analysis — Workflow

- Set up the model: building elements or masses, plus location; add analytical systems and define system zones; generate and revise analytical spaces.
- Revit translates the data to EnergyPlus, which runs the simulation — two default workflows: system sizing (heating/cooling loads) and annual energy simulation.
- Re-running with the same Building Type, Operating Schedule and Location groups results in the same folder; change them and a new folder is created.



© 2026 Tertiary Infotech Academy Pte Ltd

Reviewing Analysis Results

- Reports: Project Browser ▸ Reports ▸ Analysis Reports — a time-stamped report per run; switch Report Style between Detailed and Loads.
- Compare runs: select multiple simulations and click Compare in Results & Compare.
- Peak heating, cooling and airflow demands are also set as properties on each analytical space — the numbers live on the model.



© 2026 Tertiary Infotech Academy Pte Ltd

QUIZ

Check Your Understanding — BIM Application

1

What is a 4D BIM model?

A 3D model with the addition of time or scheduling data.

2

Avoiding conflicts between systems?

Stakeholders use clash detection software on the shared model — not emailed copies.

3

Design authoring is…

The process by which 3D software is used to develop a building information model.

4

Interference Check

Revit's built-in clash detection: Collaborate ▸ Coordinate ▸ Interference Check.

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL DO

Hands-On Labs — BIM Application

Lab 7

- View Templates, Schedules and Quantity Takeoffs

Lab 8

- Energy Analysis and Design Compliance Checks

—

- —

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 03 · HANDS-ON

View Templates, Schedules and Quantity Takeoffs

LAB 7

Operate BIM to produce controlled, quantified output: build a furniture-layout view with a scope box and visibility/graphic overrides and save it as a view template, then create a door schedule and a wall material takeoff, and add a calculated-value field for a preliminary cost estimate.

You'll build

A styled Floor 2 Furniture Layout view template, a sorted door schedule, and a wall material takeoff with a calculated cost column.

Tools: Autodesk Revit, Create View Template.rvt (LMS lab files)

© 2026 Tertiary Infotech Academy Pte Ltd

View Templates, Schedules and Quantity Takeoffs

1

of 6

Open Create View Template.rvt. Duplicate the Floor 2 plan view and rename it Floor 2 Furniture Layout, then apply the Main Zone scope box to set its crop region.

Project Browser ▸ Floor 2 ▸ Duplicate View · Properties ▸ Scope Box = Main Zone

2

of 6

Open Visibility/Graphic Overrides (VG). Override the Walls Cut pattern to Solid Fill light grey with light-grey lines, and the Furniture projection lines/pattern to red Diagonal Down-Small.

View ▸ Graphics ▸ Visibility/Graphics (VG)

© 2026 Tertiary Infotech Academy Pte Ltd

View Templates, Schedules and Quantity Takeoffs

3

of 6

On the Annotation Categories tab switch off Sections and set Grids to Halftone, then close VG and review the styled view.

VG ▸ Annotation Categories

4

of 6

Save the styling as a template: right-click the view ▸ Create View Template From View, and apply it to another duplicate to prove the styling is reusable.

Right-click view ▸ Create View Template From View

© 2026 Tertiary Infotech Academy Pte Ltd

View Templates, Schedules and Quantity Takeoffs

5

of 6

Create a door schedule: Schedule/Quantities, category Doors, add fields (Mark, Level, Width, Height, Count), and sort by Level — the schedule fills itself from the model.

View ▸ Create ▸ Schedules ▸ Schedule/Quantities ▸ Doors

6

of 6

Create a Material Takeoff for Walls with Material: Name and Material: Area, then add a Calculated Value field (e.g. Cost = Area × unit rate) to produce a preliminary cost estimate.

View ▸ Schedules ▸ Material Takeoff ▸ Calculated Value

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 7 · VERIFY

View Templates, Schedules and Quantity Takeoffs

✅ Test it

Applying your view template restyles a fresh duplicate in one click; the door schedule lists every door sorted by level and updates when a door is deleted; and the wall takeoff's calculated column prices the walls from live model quantities — the model is the single quantity and cost database.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 03 · HANDS-ON

Energy Analysis and Design Compliance Checks

LAB 8

Analyse design performance and check compliance: set the project location and energy settings, generate the energy analytical model, run a systems analysis (EnergyPlus), read the results and reports, and run an Interference Check between structural and architectural elements to find clashes.

You'll build

An energy analysis run with its report, plus an interference-check report of clashes between element categories.

Tools: Autodesk Revit, Revit Systems Analysis (EnergyPlus), your course model or the Autodesk sample project

© 2026 Tertiary Infotech Academy Pte Ltd

Energy Analysis and Design Compliance Checks

1

of 6

Open the sample architecture project. Set the project's geographic Location, then open Energy Settings and set the Mode and building type for analysis.

Analyze ▸ Energy Optimization ▸ Location / Energy Settings

2

of 6

Generate the energy analytical model — Revit creates analytical spaces and surfaces from the building elements.

Analyze ▸ Energy Optimization ▸ Create Energy Model

© 2026 Tertiary Infotech Academy Pte Ltd

Energy Analysis and Design Compliance Checks

3

of 6

Run a systems analysis workflow: choose 'System sizing' (design heating/cooling loads) — Revit translates the model to EnergyPlus and runs the simulation.

Analyze ▸ Energy Analysis ▸ Systems Analysis

4

of 6

Review the results: open Project Browser ▸ Reports ▸ Analysis Reports and double-click the time-stamped report; switch Report Style between Detailed and Loads.

Project Browser ▸ Reports ▸ Analysis Reports

© 2026 Tertiary Infotech Academy Pte Ltd

Energy Analysis and Design Compliance Checks

5

of 6

Select an analytical space and read its heating, cooling and airflow peak-demand properties — the same numbers as the report, attached to the model.

Analytical Space ▸ Properties

6

of 6

Run an Interference Check between Structural Columns/Framing and Walls/Floors; export the clash report and resolve one clash, then Refresh to confirm it clears.

Collaborate ▸ Coordinate ▸ Interference Check ▸ Run Interference Check

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 8 · VERIFY

Energy Analysis and Design Compliance Checks

✅ Test it

The Analysis Report opens with design heating and cooling loads for every zone, each analytical space carries its peak demands as properties, and the Interference Check lists each clash by element pair — after your fix and Refresh, the resolved clash disappears from the report.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC RECAP

Recap — BIM Application

- You can now: LO3 — Apply BIM and check the output

© 2026 Tertiary Infotech Academy Pte Ltd

04

TOPIC 04

BIM Documentation

Maintain BIM databases and information systems · documentation required for BIM · BIM e-submission requirements and standards

© 2026 Tertiary Infotech Academy Pte Ltd

ALIGNMENT: K8–K10 · A3, A9 · LO4

Key Concepts — BIM Documentation

1

Documents govern BIM

BIM standards, BEP, BIP, PEP, EIR, MIDP and the Common Data Environment (CDE) govern how project information is produced and shared.

2

The BEP is the baseline

The BIM Execution Plan records goals, roles, deliverables, level of detail and exchange protocols — agreed at project start.

3

Deliverables are agreed early

Site, massing, architectural, structural and MEP models, schedules, shop drawings, as-builts and FM data — with dates.

4

Databases need discipline

SVY21 geo-referencing, 1:1 metric scale, file/view naming formats, colour standards and federated file structures keep the database usable.

5

Worksharing maintains one truth

A central model with worksets, linked files and phases lets the whole team maintain one coordinated information system.

6

e-Submission has rules

BCA BIM e-submission: last-saved views, cover page, core information and discipline-specific requirements must all be checked.

© 2026 Tertiary Infotech Academy Pte Ltd

DOCUMENTATION · K8

Documentation Required for BIM

1

BIM Standard

The standards the project's BIM must follow.

2

BEP — BIM Execution Plan

Overall vision + implementation details the team follows throughout the project.

3

BIP — BIM Implementation Plan

How the organisation rolls BIM into its practice.

4

PEP — Project Execution Plan

The project-wide delivery plan BIM slots into.

5

EIR — Employer's Information Requirements

What information the employer requires, when and how.

6

MIDP & CDE

Master Information Delivery Plan schedules deliverables; the Common Data Environment is the single place information is shared.

© 2026 Tertiary Infotech Academy Pte Ltd

DOCUMENTATION · K8

BIM Deliverables — Agreed Early, With Dates

1

Models

Site model, massing model, architectural / structural / MEP models.

2

Model uses

Regulatory submissions, coordination and clash detection, visualisation, cost estimation.

3

Programme

Schedule and phasing programme — in BIM or spreadsheet.

4

Construction

Construction and fabrication models, shop drawings.

5

Handover

As-built model in native or open formats; data for facility management.

6

When agreed

At project start, and after the main project members are appointed.

© 2026 Tertiary Infotech Academy Pte Ltd

Producing Drawings and Specifications

Architecture

- Architectural guidelines

– Model in stages: Conceptual, Preliminary, Detailed, Construction, As-Built

– Use the correct tool per element (Wall tool, Slab tool …); set 'Type' correctly when improvising

– 2D may complement the model for elements smaller than the agreed size (e.g. < 100 mm)

– Model elements separately per storey

Structure

- Structural guidelines

– Structural consultant produces an analysis model AND a physical model (Structural BIM)

– Covers load-bearing concrete, wood and steel + non-load-bearing concrete

– Rebar and joint details in Detailed Design stage, per tool capability

– Required parameters: Type, Material, ID, Size — Type drives quantity takeoff

© 2026 Tertiary Infotech Academy Pte Ltd

Databases and Information Systems for BIM

Submission Requirements

2D Views	3D Views	Schedule
• Location Plan • Site Plan • Floor Plans • Elevations • Sections • Unit/ Area Plans • Area Plans for GFA • Landscape Plans • Landscape Section	• Part- 3D of floor plans • <any critical area that needs to be shown>	• Summary of Unit • Summary of Gross Floor Area • Summary of GFA Exempted • Summary of GFA Incentive • Site Coverage Computation • Communal Open Space Computation
All 2D Views and Schedules shall be placed on sheets		

The project information database: models, documents and structured data flowing through a common data environment.

© 2026 Tertiary Infotech Academy Pte Ltd

SINGAPORE STANDARDS · K9

Model Orientation, Site Configuration and Scale

1

SVY21 coordinates

Geo-reference the site model to SVY21 for Easting and Northing (x, y).

2

Singapore Height Datum

Elevations set from SHD 0.000 m for height (z).

3

True North

Present the site model layout in True North / real-world orientation.

4

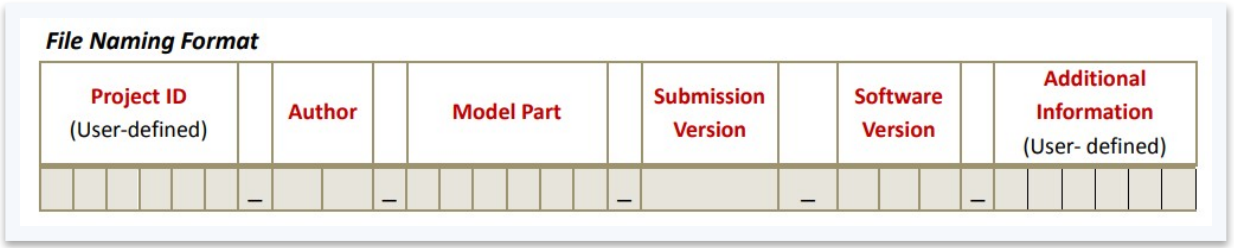
Full-size metric

Model at 1:1 in metric; 2D views generated from the model must not use odd drawing scales.

© 2026 Tertiary Infotech Academy Pte Ltd

File Naming Format

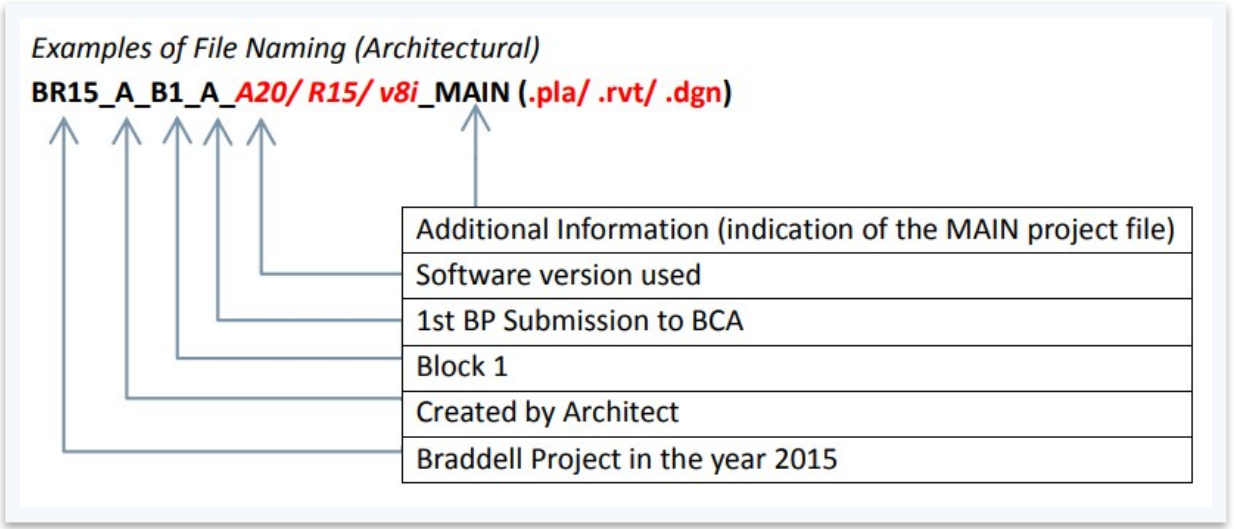
- The e-submission file name consists of 6 fields: project, originator, volume/zone, level, file type and discipline.
- Consistent naming keeps the federated model navigable for every party and every agency.



© 2026 Tertiary Infotech Academy Pte Ltd

File Naming — Worked Examples

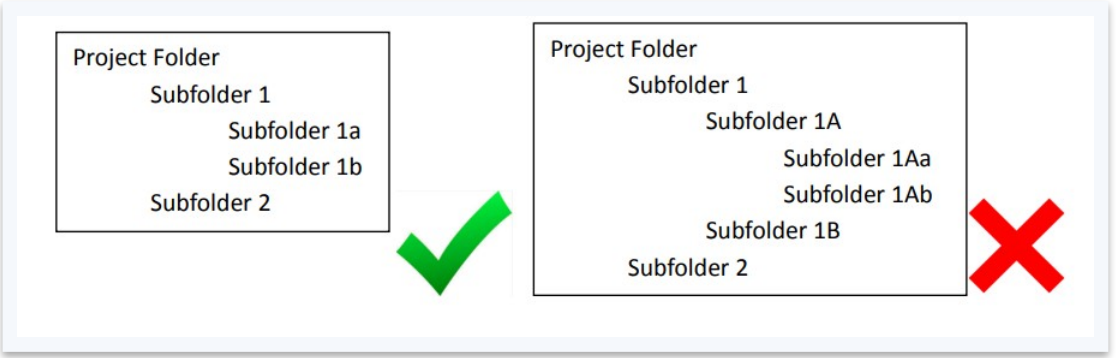
- Architectural, MEP and structural files each follow the same 6-field pattern with their discipline code.
- The examples show real project file names for each discipline.



© 2026 Tertiary Infotech Academy Pte Ltd

File Structure — Single vs Federated

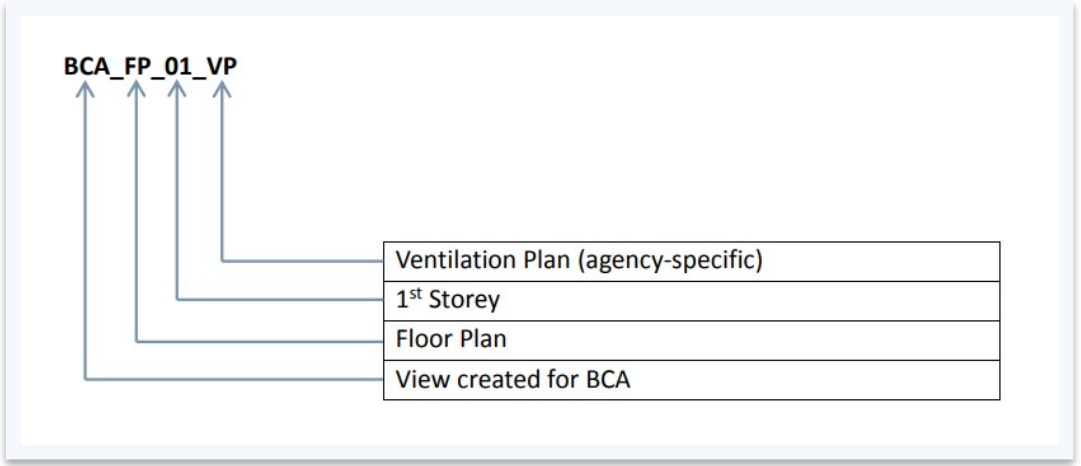
- Single file: one BIM file holds everything — suitable for small projects with one main building.
- Federated / linked files: several BIM files linked together — for multi-building or large projects.
- Use UNC or relative paths for links; folders no more than two sub-folders deep.



© 2026 Tertiary Infotech Academy Pte Ltd

View Naming Format

- View names use 3–4 fields (one field is agency-specific).
- The View Name field describes the particular view: 1st Storey, North Elevation, Section AA, Fire Protection Legend, Door Schedule.
- Views used in a submission keep exactly their View Name.



© 2026 Tertiary Infotech Academy Pte Ltd

Colour Standards for Submissions

- Amendments to approved plans use the agencies' prescribed colours.
- Addition & Alteration (A&A) works have their own colour scheme.
- Correct colours let the regulatory officer read what changed at a glance.

Colours for Amendment to Approved Plans

Colour & Line Type		RGB Values			Usage
		R	G	B	
Magenta (Solid)		255	0	255	Proposed elements
Yellow (Hidden) OR		255	225	0	Deleted elements
Orange (Hidden)		255	204	0	
Overlapping of: Magenta (Solid) and Yellow (Hidden) OR Magenta (Solid) and Orange (Hidden)					Approved elements

© 2026 Tertiary Infotech Academy Pte Ltd

e-Submission — Last Saved Views Checklist

1. **Maximum extent**
   Each view is saved at its maximum extent — agencies check the Last Saved Model and Views.

2. **Nothing hidden**
   No hidden objects or annotations remain in the submitted views.

3. **External files load**
   Every external file that forms part of the submission loads; none missing or unreadable.

4. **Purged**
   External references, irrelevant layers, drafting work and construction lines removed before submission.

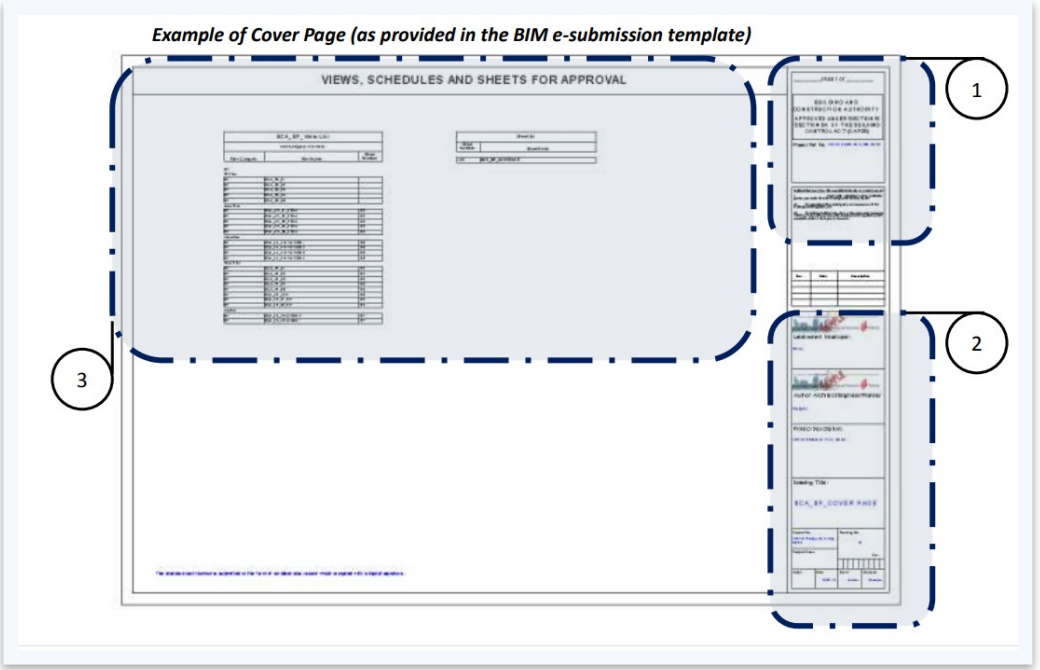
5. **Legible fonts**
   No proprietary fonts; all annotation fonts legible.

6. **All phases display**
   Objects and annotations in each phase display in the last saved view.

© 2026 Tertiary Infotech Academy Pte Ltd

e-Submission Cover Page and Core Information

- The cover page carries: (1) submission authority and endorsements / QP's declaration, (2) project information, (3) the list of views, schedules and sheets for approval.
- All submissions carry the minimal Core Information (CI) specified by the respective regulatory agency.
- Discipline-specific requirements come from the Code of Practice for BIM e-Submission.

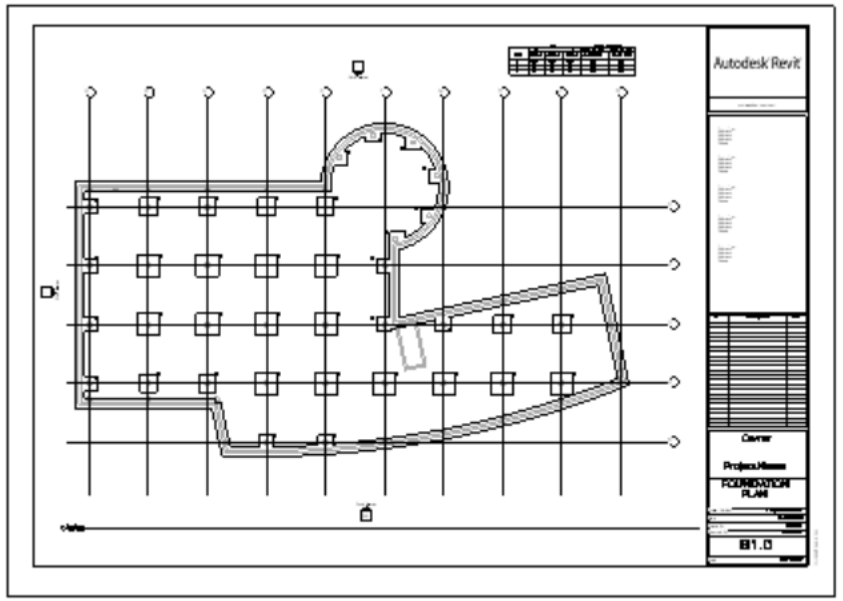


© 2026 Tertiary Infotech Academy Pte Ltd

Drawing Sheets

- A construction document set consists of sheets: View ▸ Sheet Composition ▸ Sheet.
- Drag a view or schedule onto the sheet — it becomes a viewport within the title block.
- Project templates capture sheet, view and family standards for future projects (File ▸ New ▸ Project template).

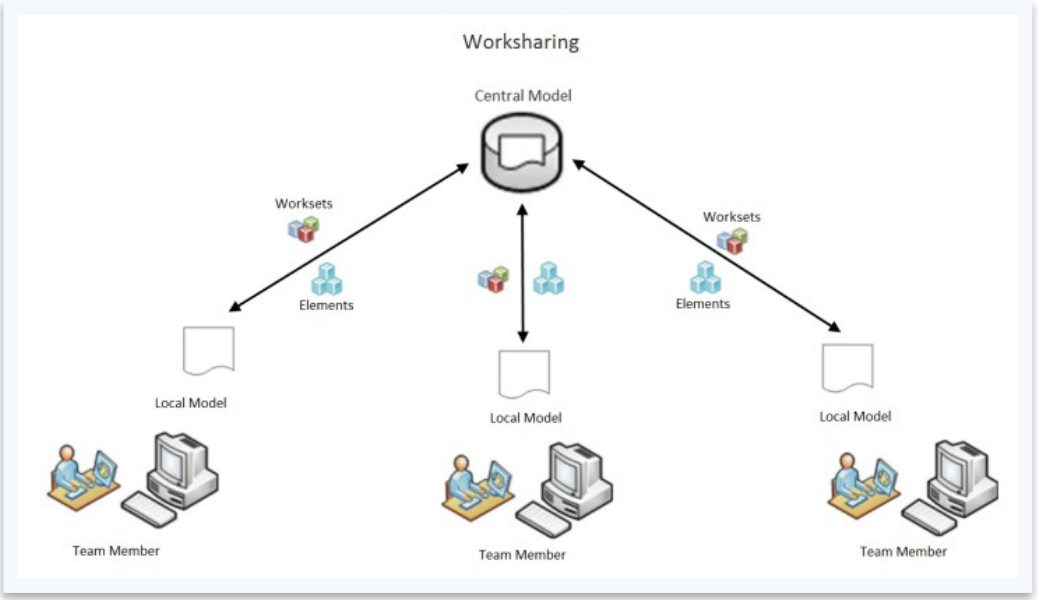
A sample sheet for a structural plan



© 2026 Tertiary Infotech Academy Pte Ltd

Worksharing — One Central Model

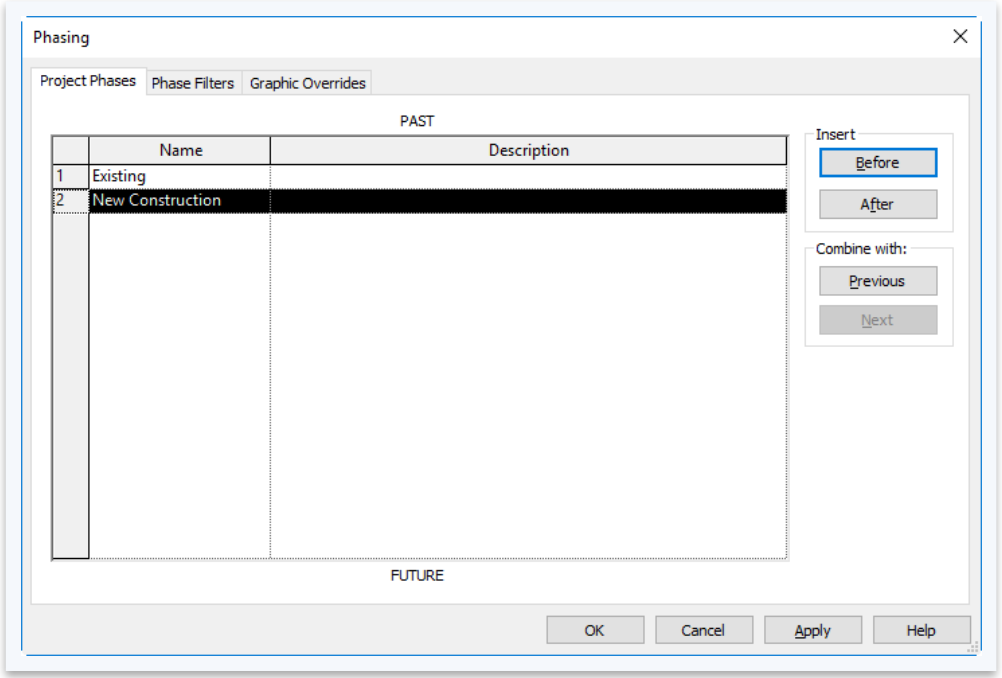
- Worksharing lets multiple team members work on the same project model at the same time, each on a functional area.
- Enable it via Collaborate ▸ Worksets: default worksets are created, then Save As publishes the central model.
- Team members edit local copies and synchronise with the central model.



© 2026 Tertiary Infotech Academy Pte Ltd

Phases and Design Options

- Phases track stages of work (Existing, New Construction, +custom): Manage ▸ Phasing ▸ Phases; insert before/after and rename.
- Design options develop alternatives (entries, roofs, layouts) inside one project file while the team continues on the main model.
- Options become more focused and simplified as the project progresses.



© 2026 Tertiary Infotech Academy Pte Ltd

QUIZ

Check Your Understanding — BIM Documentation

1

BIM Level 1

The level using 2D and 3D modeling where only CAD/PDF versions are shared with other stakeholders.

2

LOD specifications

Level of Development is defined at the BEGINNING of a project.

3

Record model

An accurate 3D representation of the physical, environmental and informational assets of a facility.

4

Where it's governed

The BEP + the agencies' e-submission Code of Practice.

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL DO

Hands-On Labs — BIM Documentation

Lab 9

- Sheets, Worksharing and Linked Models

Lab 10

- BIM e-Submission Standards Check

—

- —

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 04 · HANDS-ON

Sheets, Worksharing and Linked Models

LAB 9

Develop the construction documentation and maintain the shared model: compose a drawing sheet with a title block and viewports, add a callout and a section view, link a CAD file, enable worksharing to create a central model with worksets, and organise the project into phases.

You'll build

A titled drawing sheet carrying plan + section + callout viewports, a workshared central model with worksets, and a phased project.

Tools: Autodesk Revit, your course model, a DWG file (LMS lab files)

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 9 · STEP-BY-STEP

Sheets, Worksharing and Linked Models

1

of 6

Create a sheet with a title block and drag the Level 1 plan and a schedule from the Project Browser onto it — each becomes a viewport.

View ▸ Sheet Composition ▸ Sheet

2

of 6

Add a Section through the building and a Callout of the stair area in the plan view; open both views from their heads and place them on the sheet.

View ▸ Create ▸ Section / Callout

© 2026 Tertiary Infotech Academy Pte Ltd

Sheets, Worksharing and Linked Models

3

of 6

Link a CAD file (e.g. a surveyor's DWG) into a site view, then compare Insert ▸ Link CAD (stays external, updates) with Insert ▸ Import CAD (embedded).

Insert ▸ Link ▸ Link CAD

4

of 6

Enable worksharing: open Worksets — accept the default worksets to create the central model, then Save As the central file so team members can make local copies.

Collaborate ▸ Manage Collaboration ▸ Worksets

© 2026 Tertiary Infotech Academy Pte Ltd

Sheets, Worksharing and Linked Models

5

of 6

Create a phase: open Manage ▸ Phases, insert a phase after New Construction, and assign an element to it; set a view's Phase and Phase Filter to see the effect.

Manage ▸ Phasing ▸ Phases

6

of 6

Create a design option set for an entry alternative: Manage ▸ Design Options ▸ New, add a second option, and place variant geometry in each option.

Manage ▸ Design Options

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 9 · VERIFY

Sheets, Worksharing and Linked Models

✅ Test it

The sheet shows plan, section and callout viewports under a filled title block; the file is now a central model whose Worksets dialog lists your worksets; the phase filter hides/shows the phased element; and switching design options swaps the entry geometry — one database, fully documented.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 04 · HANDS-ON

BIM e-Submission Standards Check

LAB 10

Prepare a model for Singapore BCA BIM e-submission: apply the standard file and view naming formats, verify geo-referencing (SVY21 / SHD) and 1:1 metric scale, check the last-saved views, and assemble the submission cover page with the core information required by the regulatory agencies.

You'll build

A submission-ready model: compliant file/view names, correct coordinates and scale, purged views and a complete cover page checklist.

Tools: Autodesk Revit, your course model, BCA Code of Practice for BIM e-Submission

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 10 · STEP-BY-STEP

BIM e-Submission Standards Check

1

of 6

Rename the model file to the 6-field e-submission format (project, originator, zone, level, type, discipline) and rename key views to the 3–4 field view naming format (e.g. A_1st Storey).

File naming: <Project>-<Originator>-<Zone>-<Level>-<Type>-<Discipline>

2

of 6

Verify the site model is geo-referenced to SVY21 for Easting/Northing and to Singapore Height Datum (SHD 0.000 m) for elevation, with the layout in True North orientation.

Manage ▸ Coordinates / Position ▸ True North

© 2026 Tertiary Infotech Academy Pte Ltd

BIM e-Submission Standards Check

3

of 6

Confirm the model is built full-size 1:1 metric, and that no 2D view generated from it uses an odd drawing scale.

Properties ▸ View Scale

4

of 6

Check every last-saved view: maximum extent saved, no hidden objects or annotations, all external files loadable, irrelevant layers/drafting purged, no proprietary fonts.

Manage ▸ Purge Unused · VG ▸ Reveal Hidden Elements

© 2026 Tertiary Infotech Academy Pte Ltd

BIM e-Submission Standards Check

5

of 6

Apply the colour standards for amendments and A&A works where the submission includes changes to approved plans.

6

of 6

Assemble the cover page: submission authority and QP's declaration/endorsements, project information, and the list of views, schedules and sheets for approval, with the agency's minimal Core Information.

Sheet: Cover Page ▸ QP declaration · Project info · View list

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 10 · VERIFY

BIM e-Submission Standards Check

✅ Test it

Walking the e-submission checklist end-to-end passes: compliant file and view names, SVY21/SHD coordinates, 1:1 metric scale, clean last-saved views with nothing hidden or missing, correct amendment colours, and a cover page carrying the declaration, project information and view list — the model is ready for regulatory e-submission.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC RECAP

Recap — BIM Documentation

- You can now: LO4 — Maintain BIM database and develop BIM documentation

© 2026 Tertiary Infotech Academy Pte Ltd

WRAP-UP

Course Summary & Next Steps

© 2026 Tertiary Infotech Academy Pte Ltd

LEARNING OUTCOMES

What You Achieved

1

Applied Revit for BIM

Navigated the Revit BIM environment and turned 2D information into an intelligent 3D model (LO1).

2

Developed & integrated models

Modelled walls, floors, roofs, stairs, structure, site and masses — with families and materials (LO2).

3

Applied BIM & checked output

Built schedules and takeoffs, ran energy analysis and interference checks on the model (LO3).

4

Maintained database & documentation

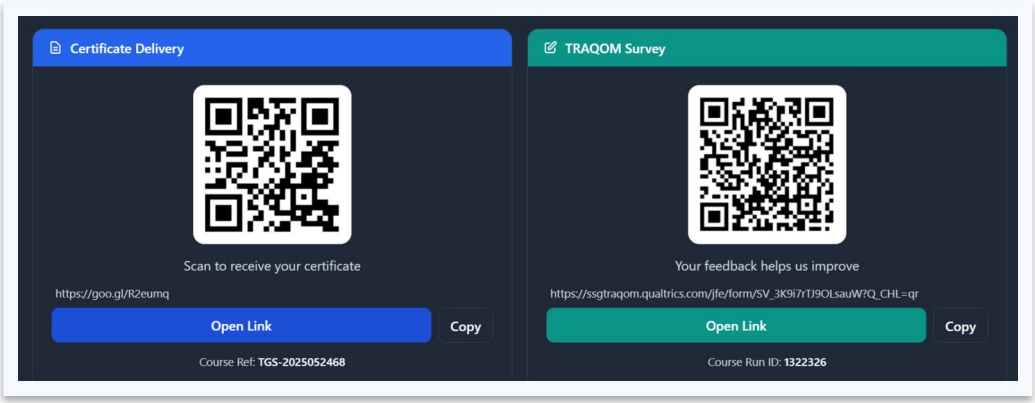
Workshared a central model, produced sheets, and met BCA e-submission standards (LO4).

© 2026 Tertiary Infotech Academy Pte Ltd

AFTER THE COURSE · lms-tms.tertiaryinfotech.com

Certificate & TRAQOM Survey (Mandatory)

- Complete the TRAQOM course-quality survey on the LMS — it is mandatory for WSQ funding.
- Your WSQ Statement of Attainment (SOA) is issued after you are assessed as Competent.
- Download your certificate of completion from the LMS portal.



© 2026 Tertiary Infotech Academy Pte Ltd

KEEP LEARNING

Recommended Next Courses

1. **WSQ - Technical Drawing with AutoCAD**
2. **WSQ - Architecture Drawing with Revit**
3. **WSQ - Product Design with Fusion 360**
4. **WSQ - AutoCAD Civil 3D for Infrastructure Design**
5. **WSQ - Product Design with Autodesk Inventor**
6. **Browse all courses**
   www.tertiarycourses.com.sg

© 2026 Tertiary Infotech Academy Pte Ltd

WE'RE HERE TO HELP

Support

- If you have any enquiries during and after the class, contact us:
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg

© 2026 Tertiary Infotech Academy Pte Ltd

FINAL ASSESSMENT

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on Revit modeling tasks, 1.5 hours, open book.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

© 2026 Tertiary Infotech Academy Pte Ltd

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ, 1 hr) then PP (1.5 hrs) — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

© 2026 Tertiary Infotech Academy Pte Ltd

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

© 2026 Tertiary Infotech Academy Pte Ltd

HAPPY MODELLING

Thank You!

You are now ready to apply, model, manage and document BIM projects with Autodesk Revit.

© 2026 Tertiary Infotech Academy Pte Ltd